

Received 27 October 2025, accepted 10 November 2025, date of publication 13 November 2025, date of current version 1 December 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3632515

SURVEY

A Survey on Task Offloading Decision-Making Algorithms in Heterogeneous Vehicular Network

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This work was supported in part by the Scientific Research Initiation Project for High-Level Talents in 2024 under Grant FWKQJ202461.

ABSTRACT Task offloading decision-making plays a crucial role in improving system performance in heterogeneous vehicular networks. This paper provides a systematic review of current research in the field and identifies limitations in existing taxonomy frameworks. It introduces a novel five-category algorithm taxonomy system, further detailing sub-categories for hybrid algorithms. This study examines algorithm performance and computational trade-offs to elucidate the characteristics and suitable application scenarios of different algorithms. It also identifies deficiencies in current research, such as the absence of standardized performance indicators and insufficient analysis of correlations among key factors. Based on research trends, future directions are proposed, including improving the performance indicators system, establishing key factor correlation models, and optimizing algorithm coordination mechanisms. This survey serves as a foundational reference for future research and paves the way for the practical application of advanced task offloading algorithms in real-world heterogeneous vehicular networks.

INDEX TERMS Heterogeneous vehicular network, task offloading, decision-making, algorithms.

I. INTRODUCTION

Intelligent transportation systems have garnered increasing attention. The Heterogeneous Vehicular Network (HVNet) integrates various communication and computing resources to construct a multi-level, complex, and efficient network environment, including vehicle nodes, roadside infrastructure, cloud servers [7]. Tasks such as autonomous driving and real-time data processing are becoming increasingly complex, placing greater demands on computing, storage, and network resources [1], [10]. However, the onboard resources of vehicles are limited. Traditional processing algorithms are challenged to meet real-time and efficiency requirements, particularly in terms of computing speed and resource utilization [2], [6]. To address these limitations, Task Offloading Decision-Making (TODM) algorithms are crucial for enhancing HVNet performance. They can efficiently match tasks with appropriate resources, optimize offloading paths and sequences, enhance resource utilization, and reduce latency [1], [12].

The associate editor coordinating the review of this manuscript and approving it for publication was Kaige Qu^{ID}.

Currently, research on TODM algorithms faces several challenges. For example, the focus varies significantly across different performance indicators. While task execution latency and energy consumption are top priorities, research into indicators like resource utilization remains insufficient and lacks unified standards [17]. This imbalance impedes a comprehensive evaluation and optimization of these algorithms. Second, in terms of algorithm characteristics, traditional algorithms and heuristic algorithms, etc., have different performances in convergence, stability, and computational complexity, making it difficult for them to adapt to the complex environment in HVNet [19]. Third, existing research shows an imbalance in the consideration of key factors such as computing and storage resources. Based on our analysis of 150 papers, the coverage of computing resources in existing research is as high as 97.3%, while the coverage of storage resources and network heterogeneity is only approximately 30%, and the coverage of vehicle mobility is as low as 16.5%.

Furthermore, existing research lacks an in-depth analysis of the correlations between various factors. For instance, few studies have explored the relationship between computing resources and network heterogeneity [39]. This gap

impedes the development of algorithms capable of adapting to real-world scenarios. Finally, in the field of computational trade-offs, research on trade-offs in resource allocation between various algorithms is insufficient. Each type of algorithm has distinct advantages and disadvantages in this context [12], [15]. Dynamic network environments and system performance fluctuations further challenge the algorithm performance [58], [72].

To obtain relevant materials on the HVNet TODM algorithm, we systematically collected academic papers. We retrieved papers published between 2014 and 2025 from platforms including Web of Science, IEEE Xplore, and ScienceDirect. During this period, HVNet technology develops rapidly, and its research achievements are representative. We used a combination of keywords including “heterogeneous,” “vehicular network,” “task offloading,” “decision-making,” and “algorithm” for the search, and initially screened out more than 200 duplicate papers. We then conducted further screening, specifically focusing on studies related to the HVNet TODM algorithm. We excluded approximately 300 papers on general vehicle networks or those not focused on decision-making. We retained empirical, experimental, and theoretical papers, while excluding approximately 150 papers that were purely conceptual. We prioritized high-impact journals and well-known conference papers in computer science, communication, and intelligent transportation, and excluded approximately 250 papers from low-impact platforms. According to the standards, we exclude more than 100 papers with low relevance to the core theme, approximately 80 papers with defective research methods, and approximately 70 papers with broad survey scopes. After multiple rounds of screening, approximately 150 high-quality papers are finally selected for the survey, laying a foundation for studying the current situation, problems, and development directions of this field.

Existing research on HVNet TODM algorithms has several deficiencies, including incomplete taxonomy of hybrid algorithms and insufficient analysis of key factor correlations. The existing taxonomy frameworks for hybrid algorithms have limitations. Specifically, the taxonomy dimension is single, and it is mostly divided according to the combination of algorithm types, without fully considering dynamic coordination. In practical applications, the coordinated operation of algorithms is essential. For instance, when traditional optimization and machine learning algorithms are combined without dynamic coordination, the algorithm response can become slow in high-speed vehicle scenarios. This, in turn, leads to a decrease in the task completion rate. In addition, the existing framework also has the problem of the lack of cross-layer optimization. HVNet contains multiple levels, and the existing taxonomy does not fully consider the coordinated optimization between different levels, resulting in low resource utilization. In terms of the analysis of the correlations between key factors, most existing research remains on the surface. Consider the correlation between the task type and vehicle mobility as an example. In actual scenarios, when

vehicle mobility is high, network instability makes it difficult for real-time interactive tasks to meet real-time requirements. However, existing research does not pay sufficient attention to this issue, which urgently needs to be addressed.

This study proposes a novel taxonomy framework for hybrid algorithms. This framework classifies algorithms into five main types: traditional optimization, heuristic, game theory-driven, machine learning, and hybrid algorithms. It also conducts a more detailed taxonomy of hybrid algorithms, including the combination of traditional optimization and machine learning algorithms, the combination of game theory-driven algorithms and heuristic algorithms, and other types. This framework comprehensively covers various algorithms and provides a new perspective and method for future research.

Table 1 compares this survey with other survey articles. [4] focuses on mobility-aware task offloading in mobile networks, which refers to the ability of offloading strategies to adapt to user mobility while maintaining system resilience. [5] focuses on Mobile Edge Computing (MEC) systems and tools, [2] and [6] involve task offloading related to the Internet of Things, [7] focuses on the architecture, challenges, and solutions of HVNet, [8] focuses on a single algorithm in the Internet of Things, [53] focuses on offloading based on Reinforcement Learning (RL) and Deep Learning (DL) in fog computing, [56] focuses on energy-aware offloading processing in mobile cloud computing, [57] explores DM factors, optimization techniques, and algorithm taxonomy in fog computing, and [98] studies edge cloud offloading algorithms. The research scopes and environments of these studies are not as precisely focused on the HVNet TODM algorithm as this survey is. In terms of research on hybrid algorithms, most of the existing surveys have an incomplete taxonomy of emerging hybrid algorithm types, only involving common hybrid algorithms, while the taxonomy framework in this study can more comprehensively and deeply explore the characteristics of hybrid algorithms. Additionally, this study comprehensively covers and explores the influence of the correlations between key factors on the algorithms. Research on key characteristics deeply analyzes the convergence, stability, and computational complexity of various algorithms in different scenarios. Research on computational trade-offs deeply analyzes the performances of different algorithms and constructs a framework to reveal the trade-off patterns. In addition, it emphasizes the standardization of performance indicators and the importance of establishing unified standards, while also providing a detailed comparative analysis of the performance changes of various algorithms in dynamic scenarios, and proposes directions for improvement.

The main contributions of this survey are as follows:

- **Proposing an innovative hybrid algorithm taxonomy framework:** Task offloading decision-making algorithms are categorized into five major types, with hybrid algorithms further divided into eight subcategories, addressing the limitations of existing taxonomies

TABLE 1. Comparison of this survey with related works in terms of research scope and coverage areas.

Papers	Research Scope	ResE	HybridA	ReleKF	KeyC	CompuTO	PerfMS	DynC
The paper	TODM Algorithms	HVNet	C	C	C	C	C	C
[2]	Single Algorithm	Internet of Things	P	C	P	P	N	P
[3]	Multi-objective Offloading Algorithms	Cloud Computing	P	P	P	C	N	P
[4]	Mobility-aware Resilient Offloading	Mobile Networks	P	C	P	P	N	P
[5]	Systems and Tools	MEC	N	P	P	P	N	P
[6]	Task Offloading and Management	IoT-Fog-Cloud	P	C	P	P	N	P
[7]	Architectures, Challenges and Solutions	HVNet	N	P	P	N	N	P
[8]	Single Algorithm	Internet of Things	N	P	P	P	N	P
[53]	RL and DL-based Offloading	Fog Computing	P	P	P	P	N	P
[56]	Energy-aware Offloading Processing	Mobile Cloud Computing	N	P	P	P	N	P
[57]	TODM Factors, Optimization Techniques and Algorithm Taxonomy	Fog Computing	N	P	P	C	N	P
[98]	Edge Cloud Offloading Algorithms	Edge Cloud	N	P	P	C	N	P

Note: C: indicates comprehensive consideration; P: indicates partial involvement; N: indicates no involvement; ResE: Research Environment; HybridA: Hybrid Algorithms; ReleKF: Relevance of Key Factors; KeyC: Key Characteristics; CompuTO: Computational Trade-offs; PerfMS: performance indicators Standardization; DynC: Dynamic Characteristic

such as their singular dimension and lack of dynamic coordination.

- **Conducting in-depth analysis of key factors and algorithm characteristics:** Through bibliometric analysis, the survey reveals the imbalance in research focus on key factors, systematically identifies the lack of correlation analysis among factors for the first time, and evaluates algorithm performance across three dimensions: convergence, stability, and computational complexity.
- **Establishing a computational trade-off and dynamic analysis framework:** A comprehensive analytical framework encompassing performance indicator, resource allocation, and algorithm efficacy is developed, revealing trade-off patterns across core dimensions, with particular focus on algorithm performance variations and adaptability in dynamic environments.
- **Identifying four key future research directions:** The survey systematically proposes critical development paths including improving the performance indicators system, establishing key factor correlation models, optimizing dynamic coordination mechanisms for hybrid algorithms, and exploring innovative algorithm combinations integrated with emerging technologies.

The subsequent sections of this paper are structured as follows: First, we present an overview of the HVNet architecture and fundamental concepts of TODM, along with the proposed algorithm taxonomy. This is followed by an in-depth analysis of four core algorithm categories and their hybrid variants. Subsequently, we conduct a systematic analysis of literature statistics and comparative studies across multiple dimensions, including performance indicator and computational efficiency. To address existing limitations, we propose future research directions such as improving the performance indicators system and establishing key factor correlation models.

Finally, we conclude by summarizing the research value and development prospects.

II. OVERVIEW

This section focuses on the content related to TODM in HVNet. First, the network architecture is presented. It then explains the relevant content regarding the TODM. Finally, the algorithms are classified and compared to prepare for a subsequent discussion.

A. HETEROGENEOUS VEHICULAR NETWORK

HVNet features a multi-tiered, distributed architecture composed mainly of vehicle nodes, roadside infrastructure, cloud servers, and communication networks, as shown in Fig. 1. Vehicle nodes collect their own status information, such as speed, location, and parameters related to vehicle health (e.g., engine status and battery level), as well as data from the surrounding environment perception, such as road conditions and traffic signals, and conduct preliminary processing and analysis. In-vehicle terminals are equipped with sensing, communication, computing, and storage functions [95].

Roadside infrastructure, including Edge Servers (ES), base stations, and traffic signal controllers, etc. [95], can provide broader communication coverage and stronger computing capabilities. It can process and integrate the data uploaded by vehicle nodes, provide feedback information, or upload data to cloud servers. The ES is responsible for collecting, processing, storing, and providing services [95].

Cloud servers, with their powerful computing and storage resources, offer decision-making support and service optimization for HVNet systems and carry out global management and centralized control [95].

The communication network is responsible for data transmission, ensuring that the data can be quickly and accurately transferred between different levels [31]. Road-side units and communication devices on vehicles are used for

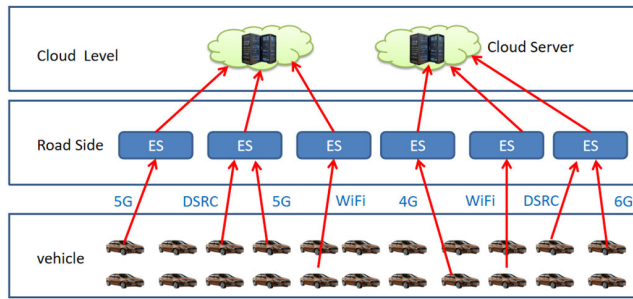


FIGURE 1. Architecture of an HVNet comprising vehicle nodes, roadside infrastructure, and cloud servers.

the exchange of task offloading information [96]. HVNet includes a wide variety of communication technologies. 4G/5G is used for large-scale data transmission. Bluetooth and Wi-Fi are suitable for short-range, low-power communication. Dedicated Short-Range Communications (DSRC) are specifically designed for vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication, featuring low latency and high reliability. HVNet includes V2V, V2I, and vehicle-to-network (V2N) communication. These communication methods, together with communication devices, form a communication system.

B. TASK OFFLOADING DECISION-MAKING

Task offloading involves transferring computing tasks from resource-constrained end devices to resource-rich remote platforms. This process aims to accelerate application execution and achieve objectives, such as reducing application execution latency [2]. The execution process comprises task partitioning, TODM, task deployment, task execution, and result retrieval. In HVNet, task offloading is suitable for complex computing tasks, big data processing, scenarios with limited resources, and applications with high real-time requirements [1], [22], [96], [97].

Task offloading methods can be classified based on various criteria. One key criterion is the feasibility of the application offloading. Application tasks can be categorized into two types: offloadable and non-offloadable. Offloadable tasks can be executed locally or offloaded to remote clouds. In contrast, non-offloadable tasks must be executed locally. This decision depends on various factors, such as network conditions, response time, and energy consumption. According to the amount of data processing, offloading is divided into full and partial offloading. Full offloading means offloading the entire task to the ES, while partial offloading allows part of a computationally intensive task to be offloaded to the edge cloud [96]. According to the offloading destination, it can be an ES, access point, or cloud. The choice of offloading destination affects the parameters of the offloading problem, and it must be selected according to the task requirements and node types. According to the dependency relationships of the offloaded parts, task offloading can be divided into different types based on factors such as task structure, optimization algorithms, and multi-user scenarios.

TODM is a complex process that requires a comprehensive consideration of multiple aspects. This process involves numerous key factors, including task, computing, storage, network heterogeneity, vehicle mobility, energy, and environmental factors. Task types are diverse, including computationally intensive, data-storage-oriented, and real-time interactive types, etc. [103]. Different task types have different requirements for computing, storage, and network resources. Tasks have several characteristics. In addition to computing requirements, data volume, and real-time requirements, they also include task priority [109], deadline, and fault tolerance, etc. The computing power requirements of different tasks vary greatly [105]. Computationally intensive tasks require high-performance processors and a large amount of computing resources to complete calculations quickly. Some simple tasks can meet these requirements with only low computing power. Data storage requirements are diverse and changeable [106]. Different types of data, such as sensor, image, and text data, etc., have different requirements for storage capacity, read-write speed, and data reliability. The limitation of the storage capacity has a significant restrictive effect. In HVNet, communication technologies show obvious heterogeneity, with multiple communication technologies such as DSRC, LTE, and 5G. In the HVNet environment, vehicle mobility is highly dynamic. There are various types of vehicles with large differences in moving speeds and uncertain moving directions [9]. When vehicles perform task offloading, energy supply and consumption are critical [51]. Different task offloading methods and decisions lead to differences in vehicle energy consumption, thus affecting the vehicle's cruising range and usage cost. The environment can cause signal attenuation and interference. Environmental factors such as temperature and humidity can affect device performance. In addition to the impact of road environment diversity on vehicle mobility, environmental factors such as weather may also affect network signals, vehicle sensor performance, etc., and thus, indirectly affect TODM [108].

The task-processing workflow is illustrated in Fig. 2. In the task-processing workflow, a task begins when it is triggered. Subsequently, the offloadability of the task must be determined, and attributes such as the computational workload, data volume, and real-time requirements of the task must be analyzed. In terms of computing resources, if local computing resources are sufficient, the non-offloadable of the task is low; otherwise, it is high. In terms of storage resources, when local storage is limited, the task tends to be offloaded on platforms with larger storage capacities to increase non-offloadable. Subsequently, the offloading time must be selected by considering the task priority and system load. The network performance is a key factor. When the network is idle, it is advisable to offload the task to reduce latency. When the network is busy, it is necessary to decide whether to wait or offload immediately according to the latency tolerance of the task. To reduce latency, priority should be given to the times when the network is

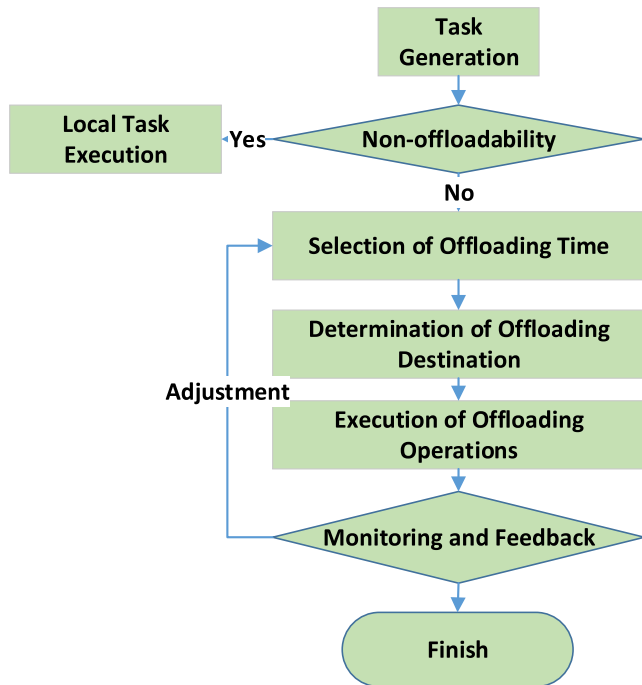


FIGURE 2. Workflow of task processing and offloading decision-making in HVNets.

good and the system load is low. When determining the offloading destination, resource statuses, such as computing power, storage, and network bandwidth of the ES and cloud servers, need to be evaluated. Computing resources affect computationally intensive tasks, storage resources are related to data storage, and network performance affects data transmission and task processing efficiency. To improve resource utilization, platforms with relatively idle resources should be selected. Subsequently, an offloading operation is performed. Task data are transmitted according to the destination and time. The network performance determines the success rate and speed of data transmission. Finally, monitoring and feedback are conducted. The offloading and execution status of the tasks are monitored in real time. The offloading strategy is adjusted based on the monitoring results, including reselecting the destination and time. Arrows indicate the feedback path. Through continuous monitoring and feedback, the offloading decision is optimized to achieve goals such as reducing latency and improving resource utilization.

TODM involves making decisions through analysis, judgment, and trade-offs among multiple choices during the task offloading process [8]. In this process, Various factors must be considered to determine whether to offload and how to achieve optimal resource allocation and performance improvement [3]. Before performing task offloading, the following decisions must be made [3]: 1) offloading content: Determine the offloading workload and specific tasks. The task scheduler decides whether a task can be offloaded and the offloading capacity; 2) offloading time: the task scheduler selects an appropriate time period and determines the

offloading interval; 3) offloading location: if remote offloading is chosen, the best ES needs to be selected from available nodes; 4) offloading method: select the offloading channel or path; 5) offloading strategy: clarify the offloading goals, such as optimizing a single performance indicator, joint optimization, or multi-objective trade-offs.

In the HVNet environment, the optimization goals of task offloading algorithms mainly include the following aspects: 1) Performance optimization, which includes reducing latency, increasing throughput, and enhancing reliability; 2) Resource utilization efficiency, which involves optimizing computing and communication resources and improving energy efficiency [97], [101]; and 3) Cost control, which encompasses economic cost, time cost, and quality of service [100], [102].

C. ALGORITHM TAXONOMY

In HVNet, TODM is crucial for the system performance. HVNet integrates various communication and computing resources to construct a multi-level network. Owing to the complexity of this environment, TODM algorithms have become key to improving resource utilization and reducing task latency. According to their core principles, TODM algorithms can be classified into the following categories (Fig. 3):

1) Traditional optimization algorithms: These are based on precise mathematical models to optimize specific single objectives, including linear, nonlinear, and dynamic programming.

2) Heuristic algorithms: They rely on heuristic rules and local search to find near-optimal solutions, such as genetic algorithms and simulated annealing algorithms.

3) Game theory-driven algorithms: Using game theory, they consider the relationships among participants and are divided into noncooperative and cooperative games.

4) Machine learning algorithms: They adjust strategies through learning from a large amount of data and include supervised learning, unsupervised learning, and reinforcement learning.

5) Hybrid algorithms: They combine the advantages of different algorithms, including:

(1) Hybrid of traditional optimization and machine learning algorithms: This combines the precision of traditional algorithms and the adaptability of machine learning algorithms.

(2) Hybrid of game theory and heuristic algorithms: This integrates the analytical advantages of game theory and the search ability of heuristic algorithms.

(3) Hybrid of traditional optimization and heuristic algorithms: This combines the systematic nature and precision of traditional algorithms with the flexibility and fast-solving ability of heuristic algorithms.

(4) Hybrid of game theory and traditional optimization algorithms: It uses game theory to build models, combines traditional optimization algorithms for solutions, and introduces a dynamic adjustment mechanism.

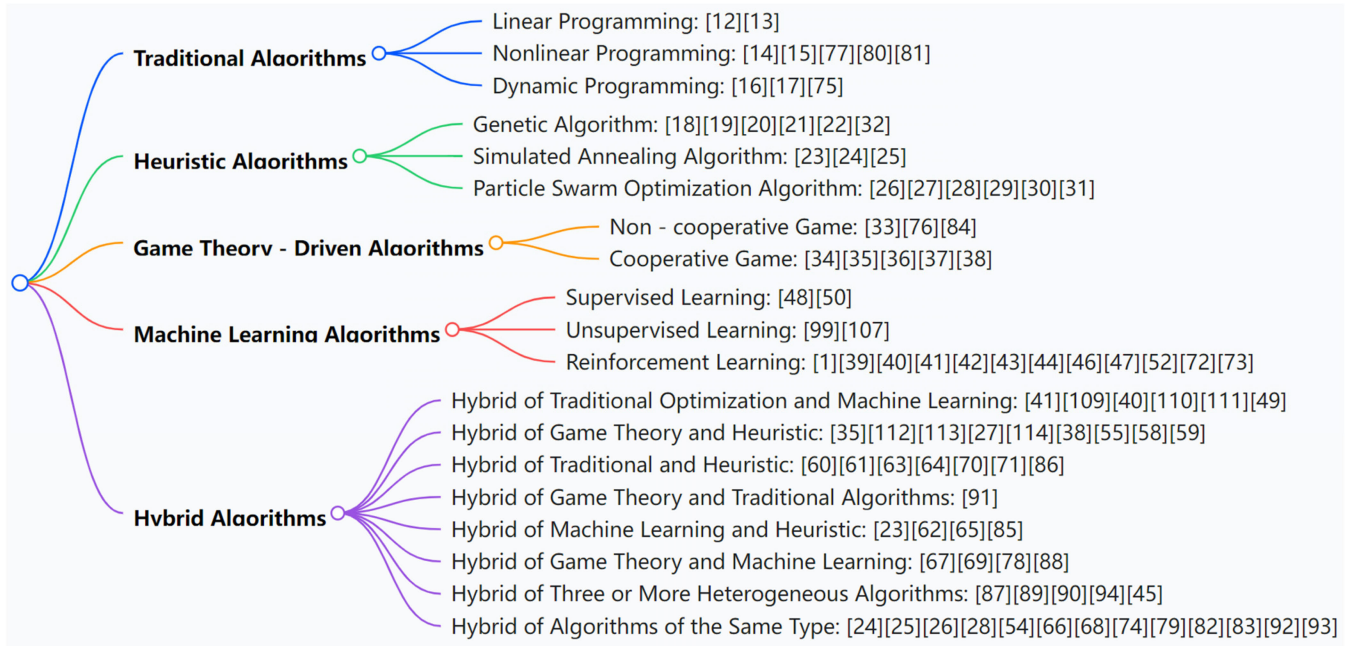


FIGURE 3. Taxonomy of task offloading decision-making algorithms in HVNet.

(5) Hybrid of machine learning and heuristic algorithms: This gives full play to the learning and adaptation ability of machine learning algorithms and the high efficiency of heuristic algorithms.

(6) Hybrid of game theory and machine learning algorithms: It constructs a multi-agent system, where game theory analyzes benefits and machine learning algorithms optimize strategies.

(7) Hybrid of algorithms of the same type: This combines different algorithms of the same category.

(8) Hybrid of multiple heterogeneous algorithms: This combines multiple types of algorithms.

Different TODM algorithms have their own unique characteristics. In practical applications, the choice should be based on the scenarios, tasks, and system resources. Standalone algorithms exhibit limitations in complex environments, whereas hybrid algorithms demonstrate distinct advantages. With the development of technology, algorithm research continues to deepen, promoting the innovation of TODM algorithms in HVNet and meeting the needs of fields, such as intelligent transportation.

III. SINGLE ALGORITHMS

Single algorithms play a crucial role in the HVNet TODM algorithm ecosystem. Facing complex offloading scenarios, different algorithms have various characteristics, advantages, and disadvantages. This section details the principles, applications, and dynamic characteristics of each algorithm.

A. TRADITIONAL OPTIMIZATION ALGORITHMS

This type of algorithm usually solves problems based on precise mathematical models, with the aim of optimizing a

specific single objective. It includes linear, nonlinear, and dynamic programming algorithms.

1) LINEAR PROGRAMMING

Linear programming aims to optimize a linear objective function subject to a set of linear constraints. In the task offloading scenario of HVNet, factors such as the local computing resources of vehicles, computing capabilities of edge and cloud servers, communication bandwidth, and task priorities must be comprehensively considered to construct a function. Then, methods such as the simplex method or the interior-point method can obtain an accurate task offloading scheme.

Amani et al. [12] found that in an actual network, the capacity of access points changes owing to the number of users and service types, and the energy consumption cost is affected by energy prices and device status. The optimal resource-allocation scheme at a specific moment can be obtained by constructing a linear programming model and incorporating these dynamic constraints. The computational complexity of this model is low. Although its ability to handle complex scenarios is limited, this approach can quickly provide a satisfactory solution based on current conditions and effectively manage resource allocation dynamics at a reasonable cost.

Mukherjee et al. points out that in a heterogeneous fog computing environment, the aging of fog node devices, uneven loads, network traffic, and changes in user requests cause fluctuations in energy consumption and latency [13]. By constructing a linear programming model and incorporating these dynamic constraints, the optimal TODM and resource-allocation scheme can be determined. Linear programming has a high solution efficiency, can quickly respond

to dynamic changes, and balances computational efficiency and adaptability.

2) NONLINEAR PROGRAMMING

Nonlinear programming is used to solve the task function and constraint conditions. In HVNet, the latency is affected by various factors and exhibits a nonlinear relationship. So, when constructing the model, these relationships must be accurately described, and then a suitable algorithm is selected for the iterative solution.

Regarding resource allocation dynamics, Xu et al. addresses a scenario where energy consumption is affected by various factors, resulting in a nonlinear objective function, and the resource requirements and the environment are constantly changing [14]. Therefore, they employed a matching game and convex optimization. The matching game is used to dynamically match resources with demands, and convex optimization is used to solve the optimal resource-allocation scheme for the changes. This method is computationally complex, but it can balance resource allocation and energy consumption, and has a stronger adaptability to complex dynamic environments compared to simple algorithms. Fan et al. set the task offloading and resource allocation as a mixed-integer nonlinear programming problem and uses a branch-and-bound algorithm improved by linear relaxation to solve it [15]. However, this method faces challenges, such as a difficult solution, high computational complexity, and being prone to falling into a local optimum.

In terms of problem transformation and dynamic adaptation, Wang et al. uses methods such as binary variable relaxation and mathematical function substitution to transform the original nonconvex problem into a convex problem [77]. Because dynamic changes in the actual scenario may change the nature of the problem, a mature convex optimization algorithm can be used for the solution after the transformation. Although the transformation increases the computational cost, it can quickly adapt to the changes with the help of efficient algorithms in a dynamic environment and balance the computational cost and dynamic adaptability.

For multi-objective dynamic optimization, Liu employs the Interior Point Method (IPM) [80]. In practice, multiple objectives such as resource utilization, cost, and task completion quality change dynamically over time. The complexity of IPM is a low-order polynomial, and it can provide an optimal solution within an almost constant number of iterations. By introducing a penalty function to deal with dynamic constraints, it performs excellently in efficient calculation and adaptation to the dynamic changes of multiple objectives and realizes multi-objective dynamic optimization at a low cost.

In terms of dynamic task scheduling, Yang proposes a low-complexity maximum-energy-efficient task scheduling (MEETS) algorithm to address fluctuations in task volume, priority, and node status within a homogeneous fog network energy-efficient scheduling framework [81]. Specifically, the algorithm models the problem using a multi-node

fractional programming framework and iteratively refines the task scheduling and resource allocation decisions. Although the solution is complex and the computational complexity is high, it can effectively deal with the dynamics of task scheduling and improve the energy efficiency.

3) DYNAMIC PROGRAMMING

Dynamic programming decomposes the TODM problem into multiple interrelated subproblems. The stage and state variables are determined according to the driving state of the vehicle, network changes, etc., and a recurrence relationship is constructed. Starting from the initial state, each subproblem is solved step by step, and finally, the overall optimal decision-making strategy is obtained.

This approach has been applied in fog computing systems. For example, Li et al. employs dynamic programming to buffer tasks and optimize task offloading/resource allocation in stages, addressing the variability in task urgency, arrival time, and computing requirements [17]. Although the computational cost increases with the number of stages and states, this approach enables accurate decisions, improves real-time task processing, and optimizes system performance—albeit at a high computational cost. In a complex system environment, with the help of Lyapunov optimization to decompose the problem, such as dynamic programming, Qin et al. solves it step-by-step according to the real-time state [16]. Although this increases the computational cost, it reduces the complexity, balances the computational cost and dynamic adaptability, and helps optimize task scheduling and resource allocation. In an MEC system, based on Lyapunov optimization, Mao et al. proposes an online algorithm to deal with dynamic changes, such as the task load of multiple devices [75]. This algorithm updates the strategy in real time. Although it consumes computational resources, it can balance the power consumption and task buffer stability, provide a supplement for improving the system performance, and help the system operate efficiently in a dynamic environment.

B. HEURISTIC ALGORITHMS

This type of algorithm relies on heuristic rules and local search strategies to find near-optimal solutions to problems quickly. It includes genetic algorithms, simulated annealing algorithms, and particle swarm optimization algorithms, etc.

1) GENETIC ALGORITHMS

Genetic algorithms simulate biological evolution. In this process, the algorithm encodes the task offloading scheme as a chromosome, where each chromosome represents a potential solution. A fitness function is used to evaluate the quality of chromosomes. Chromosomes with a high fitness are more likely to be selected for reproduction in the next generation. During reproduction, some genes are exchanged through crossover, and some genes are randomly changed through mutation, promoting the evolution of the population to approach the optimal solution.

Chen et al. finds in his research that in an edge cloud server environment, task dependencies, resource conditions, and energy consumption requirements change at any time [19]. To this end, he designs a TDM scheme based on a genetic algorithm and improved the algorithm using a skewed mutation operator. This scheme operates on chromosomes according to the task dependencies and system constraints in each iteration. Although the computational cost increases, it can accurately deal with dynamic changes and effectively balance the computational cost and optimization effect. Related to this, Sundar et al. uses a genetic algorithm to find the optimal offloading strategy for the dynamic changes in HVNet [18]. Although the computational cost is large, it can update the search direction according to changes in the system; Chakraborty and Mazumdar makes TODM from multiple aspects with the help of a genetic algorithm in the sensor MEC scenario and optimizes the energy requirements by adjusting the evaluation criteria according to dynamic information [20]; Materwala et al. adopts a genetic algorithm combined with an adaptive penalty function to find the offloading scheme with the minimum energy consumption in the task offloading scenario of HVNet edge cloud integration, which can adapt to changes in real time [21]; Zhu and Wen proposes an offloading strategy based on an improved genetic algorithm in the MEC system and uses integer encoding to enhance the optimization ability, flexibly deal with dynamic changes, and improve system adaptability [22].

In terms of multi-objective dynamic optimization, Huynh et al. notes that in multi-objective task offloading, both the objectives and the system state change [32], making it necessary to adjust the fitness function—a process that is difficult to achieve. When the state changes, the function parameters must be readjusted, resulting in a large computational cost and time consumption. Careful design can optimize multiple objectives in a dynamic environment and achieve balanced optimization of multiple objectives at a high cost.

2) SIMULATED ANNEALING ALGORITHMS

The simulated annealing algorithm is based on the physical annealing process. It starts from a relatively high “temperature” state and accepts solutions that are worse than the current solution with a certain probability. As the “temperature” gradually decreases, the probability of accepting worse solutions also gradually decreases, and finally converges to a near-optimal solution. This mechanism helps the algorithm escape local optima and explore better offloading schemes by continuously adjusting the state of the solution.

Regarding dynamic task scheduling and offloading, Mahjoubi et al. [23] highlight that in a hierarchical MEC architecture, the states of tasks and resources are changeable. To achieve efficient online task scheduling, he introduces the SATS (Simulated Annealing-based Task Scheduling) algorithm based on simulated annealing. This algorithm can adjust the decision making in real time according to the system state. Although the iterative computational cost increases

with the number of tasks and resources, it can find a better solution in a dynamic environment and balance computational cost and dynamic adaptability. Yuan et al. points out that the states of tasks and resources change significantly in a hybrid cloud-edge system [24]. The SMBO (Simulated Annealing-based Migrating Birds Optimization) algorithm based on simulated annealing can solve the task offloading problem of profit maximization. When the algorithm is executed, the computational cost is relatively large, but it can explore new solutions in a dynamic environment, jump out of the local optimum, and search for the optimal offloading solution at a high computational cost.

In terms of multi-objective dynamic optimization, Bi et al. states that for single-objective task offloading, the simulated annealing algorithm converges to a better solution by adjusting the temperature parameter, and the computational cost is controllable. For multi-objective cases, owing to the need to balance multiple objectives, adjusting the parameters increases the computational cost [25]. After reasonable adjustment, it can optimize multiple objectives in a dynamic environment and balance the computational costs and effects. Shu and Li mention that in task offloading, the states of tasks and resources change dynamically [28]. The simulated annealing algorithm widely explores the solution space at a high temperature and offloads tasks according to the requirements. Although the computational cost is large, it can adapt to changes and increase the possibility of finding a good solution in a dynamic environment at a high computational cost, balancing dynamics, and computational cost.

3) PARTICLE SWARM OPTIMIZATION ALGORITHMS

The particle swarm optimization algorithm simulates the foraging behavior of bird flocks. Each particle is regarded as a potential solution in the solution space. Particles have two attributes, position and velocity, and they adjust their velocity and position according to their own historical best and the group’s best position to search in the solution space. In the task offloading of HVNet, the position of the particles can represent the task offloading scheme, and the particles can quickly find a better offloading scheme through information sharing and collaboration.

Regarding dynamic task offloading and resource allocation, Liu et al. addresses a scenario involving dynamic online task offloading, where the workflow, network topology, and computing resource availability undergo constant change [29]. The author employs a particle swarm optimization algorithm to jointly optimize workflow allocation and multihop routing scheduling. Within this algorithm, particles dynamically adjust their positions and velocities in response to environmental fluctuations to identify near-optimal solutions. Although the computational cost increases with the number of particles and scale of the problem, information sharing among particles enables the algorithm to quickly adapt to changes and improve the reliability and efficiency of offloading at a relatively high cost. Related to this, Qiu et al. proposes a triple-check offloading algorithm (TCOA) in an

in-vehicle micro-cloud environment for situations where the states of tasks and vehicle resources are changeable [31]. By combining the gradient descent and binary search, the gradient and search range are calculated according to the dynamic information during the iteration. Although the computational cost increases with the number of tasks and vehicles, it can adjust the offloading ratio and number of vehicle instances in real time, balance the computational cost and dynamic adaptability, and optimize task offloading and resource allocation. Shu and Li introduces a joint offloading strategy based on quantum particle swarm optimization in the HVNet scenario in the face of continuous changes in task attributes and network states [28]. Search diversity increases with the help of quantum characteristics. Although the calculation is complex, it can better adapt to the dynamic changes of HVNet and optimize the system performance.

Zhang et al. notes that task execution reliability and efficiency during offloading are influenced by dynamic factors such as task urgency and resource load [27]. The positions and velocities of the particles are updated using the particle swarm optimization algorithm to search for a high-reliability offloading scheme. In the case of single-objective optimization, the algorithm quickly converges to a better solution, and the computational cost of each iteration is controllable. In a dynamic environment, the algorithm adjusts the search direction according to the changes in tasks and resources and optimizes reliability and efficiency at a low cost. Huda and Moh believes that in an actual scenario, the reliability and efficiency of task execution change owing to changes in tasks and the environment [30]. The particle swarm optimization algorithm provides a method for TODM by simulating the foraging behavior of bird flocks. During the operation of the algorithm, the particles adjust their positions and velocities according to their own and neighborhood best. Although the position and objective value must be calculated in each iteration, the algorithm is simple and easy to implement. Information sharing among particles enables it to quickly adapt to changes and improves the reliability and efficiency of task execution at a low cost.

In terms of the dynamic optimization of task execution reliability and efficiency, Zhang et al. find that during task offloading, execution reliability and efficiency are affected by factors such as task urgency and resource load [27]. He uses the particle swarm optimization algorithm to search for a high-reliability offloading scheme by updating the positions and velocities of the particles. Huda and Moh proposes that, in an actual scenario, the reliability and efficiency of task execution will change with changes in tasks and the environment [30]. By using the particle swarm optimization algorithm, particles adjust their positions and velocities according to their own and neighborhood best, which is simple and easy to implement and can improve the reliability and efficiency of task execution at a low cost.

In terms of multi-objective dynamic optimization, Dong et al. finds that the multiple objectives in the task offloading scenario will change with the task type, quantity,

and resource usage [26]. To solve this problem, he designs a fitness function and uses the particle swarm optimization algorithm to deal with multi-objective problems. In each iteration, the fitness of the particles is calculated according to the dynamic objective values and fitness function. Owing to the complexity of multiple objectives, the computational cost is relatively large, and the parameters of the fitness function also need to be adjusted to balance multiple objectives, thereby increasing the computational complexity. However, this algorithm can optimize multiple objectives in a dynamic environment and effectively balance the computational cost and optimization effect. Liu et al. proposes that in a dynamic environment, determining the destination of task offloading requires a comprehensive consideration of multiple objectives [29]. Although the particle swarm optimization algorithm is simple, easy to implement, and converges quickly, it is prone to fall into a local optimum and is sensitive to the initial values when dealing with complex multiple objectives, and there is a contradiction between the computational cost and the optimization effect.

C. GAME THEORY-DRIVEN ALGORITHMS

Applying game theory to TODM, this algorithm fully considers the competitive and cooperative relationships among different participants, such as various vehicles, ES, and cloud servers, and is divided into non-cooperative and cooperative games.

1) NONCOOPERATIVE GAME

Competition is at the core of non-cooperative games. Each participant pursues the maximization of its own interests and independently chooses the offloading destination. This approach analyzes the strategy spaces and payoff functions of all participants to determine the Nash equilibrium state. In this state, each participant believes that their current strategy is best when the strategies of other participants remain unchanged, thus achieving a relatively stable state [33].

In terms of the dynamics of TODM, when Yang et al. studies multi-user task offloading, facing complex dynamic changes, he innovatively transforms the task offloading problem into a potential game problem [84]. He solves the Nash equilibrium strategy through a dynamic analysis of the relevant factors. This process depends on real-time information updates and has a relatively large computational cost; however, it also has obvious advantages. This strategy can accurately determine the optimal offloading and resource allocation strategy according to environmental changes, effectively optimize the scheme in the heterogeneous fog network scenario, and balance the computational cost and optimization effect well. Chen et al. constructs an algorithm using noncooperative game theory, allowing users to make real-time decisions that can achieve efficient offloading and adapt to changes [76]. However, when the number of users increases, the computational cost surges, and this problem is particularly prominent in the MEC scenario.

In terms of the dynamics of system resource allocation, when vehicles offloading tasks, their computing resources, task types, and network connections constantly change. Wang et al. points out that, without external constraints, vehicles make decisions based on their own interests, which leads to an imbalance in the resource allocation of HVNet [33]. Calculating the benefits of local strategies for vehicle computing has a small computational cost and is beneficial to individuals; however, it disrupts the system's resource allocation. If the Nash equilibrium analysis is used for improvement, it is necessary to comprehensively consider the impact of the overall vehicle strategy combination, which will increase the computational complexity and highlight the contradiction between the computational cost and the dynamic optimization of resource allocation.

2) COOPERATIVE GAMES

Cooperative games emphasize win-win cooperation. In an actual operation, the cooperating parties are first determined, then the cooperation benefits are evaluated, and a reasonable cooperation mechanism is designed. In the process of resource integration and task offloading, all parties cooperate through negotiation to jointly optimize the overall objectives, achieving single- or multi-objective optimization, thus improving the task completion rate and reducing the system latency [34].

Sun et al. finds that in cooperative games, factors such as the resources of vehicles, ES, cloud servers, task requirements, and user preferences change rapidly [38]. Therefore, they propose the BARGAIN-MATCH (Bargaining and Matching) algorithm. This algorithm uses bargaining and matching schemes to allocate resources and offload tasks and achieves multi-objective optimization. During the operation, a continuous calculation is required according to the dynamic information. As the number of participating parties and complexity of tasks increase, the computational cost increases exponentially. Although balancing the interests of all parties, the algorithm mechanism is complex, and the computational cost is high, it can effectively balance the computational cost and the effect of multi-objective optimization in a dynamic environment. The cooperation mechanism and negotiation process also enhance the ability of the algorithm to deal with dynamic changes. In contrast, Hui's et al. two-stage mechanism focuses on task offloading for cooperative vehicles in HetNets [35]. This mechanism determines the minimum task completion cost of small base stations and bargains between parties to set the price of offloading services and achieve single-objective dynamic optimization at a relatively high computational cost. Zhang et al. proposes a hierarchical framework for the cloud-based Vehicle Edge Computing (VEC) offloading scenario using Stackelberg game theory to design a multilevel offloading scheme [36]. Although the computational cost is large, it can adapt to dynamic changes and achieve multiobjective optimization. Huang et al. constructs the cooperative relationship of vehicles using

cooperative games in the HVNet system and designs an iterative matching game algorithm to find the optimal offloading strategy [37]. Although the computational cost increases with an increase in the complexity of vehicles and tasks, it can improve the system performance and balance the computational cost and the effect of multi-objective optimization.

D. MACHINE LEARNING ALGORITHMS

Machine learning algorithms automatically adjust strategies through the learning and analysis of a large amount of data to adapt to complex and changeable network environments. These algorithms are mainly divided into supervised, unsupervised, and reinforcement learning.

1) SUPERVISED LEARNING

Supervised learning is trained using labeled data to learn the mapping relationship between input features, such as vehicle task types, network latency, and server load, and output labels, such as whether to offload and the offloading location. A decision-making model is constructed based on this learning process; a decision-making model is constructed [48].

For example, using the decision tree algorithm, one can construct a model based on the dynamic characteristics of vehicles to determine the task offloading strategy [48]. When constructing the model, a large amount of computational analysis of labeled data is required to determine the correlation between the features and TODM. For example, when the feature values change, the change in the vehicle's location affects the network latency, or the server's load changes owing to new tasks, the model may need to be re-evaluated and updated. In a stable environment, after the model is constructed, it can make decisions quickly, determine the task offloading strategy according to the input, and achieve single-objective optimization, such as reducing the total latency cost and increasing resource utilization. However, in the dynamic environment of HVNet, the data distribution constantly changes, and new situations occur frequently. The decision tree model may need to be frequently rebuilt and adjusted, which requires a large amount of computational resources to reprocess the data, calculate the feature importance, and construct the tree structure, resulting in a significant increase in the computational cost.

Y. Qin continuously interacts with the wireless network environment through the Online Learning to Parallel Offloading (OLPO) algorithm and adjusts the network selection, effectively reducing the task latency and solving the problem of the dynamic change of network attributes. Simultaneously, by studying the algorithm performance under different numbers of networks, parallel offloading of 2-3 networks is more efficient, providing a decision-making basis for dealing with the dynamic change of the network scale [50].

2) UNSUPERVISED LEARNING

Unsupervised learning is used to discover potential patterns and structures in data through data mining techniques without labeled data. In the task offloading of HVNet, tasks can be

classified according to their characteristics, such as computing requirements and data volume, providing a basis for resource allocation and TODM.

Although MEC can reduce the energy consumption and latency of mobile devices, the computational resource allocation is extremely large. Chen et al. proposes a multiuser MEC framework based on unsupervised deep learning [99]. This framework aims to reduce the energy consumption of mobile devices by jointly optimizing binary TODM and resource allocation. However, in actual operation, this optimization problem is likely to lead to a vanishing gradient, thereby affecting the effect. To overcome this problem, a novel binary computation offloading (BCOS) scheme is carefully designed. This scheme uses a DNN (Deep Neural Network) with an auxiliary network to obtain lossless gradient information and then obtains a suboptimal solution. Through simulation verification, BCOS effectively solves the binary offloading problem by relying on a low-complexity training network. At present, when terrestrial cellular communication networks face the demand for large-scale connections and unpredictable data traffic caused by natural disasters, network congestion is often encountered. To this end, Tsipi et al. proposes a drone-assisted terrestrial network offloading strategy using an unsupervised machine learning method, dividing the user clusters in high-traffic areas with the k-medoid algorithm, and selecting appropriate clusters for deployment according to the number of drones to achieve maximum offloading, achieving significant gains in many aspects [107].

3) REINFORCEMENT LEARNING

Reinforcement learning is achieved through the interaction between an agent (such as a vehicle) and the environment (network environment, server resource status, etc.). A series of actions is taken by the agent in the environment, and the corresponding rewards or punishments are given by the environment based on these actions. The agent learns the optimal strategy by maximizing long-term cumulative rewards. In the task offloading scenario, the agent selects the offloading strategy according to the current environmental state, obtains reward feedback based on the offloading result, and continuously adjusts the strategy to achieve optimal TODM.

In terms of dynamic task offloading/resource allocation, Pang et al. proposes a game-theoretic framework in which a base station acts as a strategic agent to optimize wireless resource allocation in HVNet [41]. Facing continuous changes in sub-channels, task traffic, etc., using the MADDPG (Multi-Agent Deep Deterministic Policy Gradient) algorithm, the agent makes decisions on subchannel allocation and power adjustment according to dynamic information. During training, the agent has many interactions with the environment, and the state and action spaces are extremely large, resulting in a huge computational cost. However, it can effectively deal with the dynamics of HVNet, optimize task offloading and resource management, and balance the computational cost and optimization effect. In contrast, Wen et al. adopts multi-agent reinforcement learning

in the MEC network, and the agents adjust their strategies according to the real-time network state [46]. The interactive calculation during training leads to an exponential increase in the computational cost to optimize task offloading and resource allocation; Tuong et al. in NOMA-assisted MEC network [47], Sun et al. in HVNet [39], and Kumar et al. in HVNet [40], respectively, for the dynamic changes in their respective scenarios, uses reinforcement learning-related algorithms. By adjusting strategies and optimizing decisions, etc., they achieve goals such as energy and latency control, optimize TODM, and minimize energy consumption and latency when the computational cost increases.

In terms of single-objective dynamic optimization, Liu et al. improves the latency performance of the VEC system through the fluctuation-aware learning-based task offloading algorithm [42], and Sun et al. improves the latency of the VEC system by using the adaptive learning-based task offloading algorithm [44]. They balance the computational cost and single-objective optimization in different ways.

In terms of multi-objective dynamic optimization, Pang et al. also uses a multi-agent deep reinforcement learning algorithm to comprehensively consider dynamic factors, such as task transmission and computing time, and optimize multiple objectives, such as system overhead and task completion rate [41]. When the agents interact, multiple objectives are optimized synchronously, and the strategy update requires comprehensive calculation of the feedback of multiple objectives, resulting in a huge computational cost. However, it can effectively balance and optimize multiple objectives in a complex dynamic environment and improve the system performance at a high computational cost. Khanh et al. TT uses fuzzy logic and the SARSA (State-Action-Reward-State-Action) learning collaboration algorithm to handle multi-objective task offloading, comprehensively considering multiple objectives in a dynamic environment, and optimizing TODM at a relatively high cost [52].

In addition, Pang et al. also points out that when the reinforcement learning algorithm is used in the complex and dynamic environment of HVNet, the state and action spaces are large, the agents learn strategies through many interactions, resulting in a large consumption of computational resources, and the design and adjustment of the reward function affect the performance of the algorithm [41]. In practical applications, it is necessary to balance the investment of computational resources and improve algorithm performance.

In summary, as shown in Table 2, these four types of single algorithms have their own advantages and limitations when dealing with TODM problems in HVNet.

Traditional optimization algorithms, such as linear, nonlinear, and dynamic programming, vary in performance. For instance, linear programming has a low computational complexity and poor adaptability in complex scenarios. Nonlinear programming can adapt to complex dynamic environments; however, it is difficult to solve and prone to becoming trapped

TABLE 2. Comparative analysis of single-category task offloading decision-making algorithms.

Algorithm Type	Sub-type	Advantages	Limitations	Application Scenarios
Traditional optimization algorithms	Linear programming	Low computational complexity and can quickly obtain suboptimal solutions	Limited adaptability to complex scenarios	Used to determine task offloading schemes, such as resource allocation in heterogeneous cloud radio access networks
	Nonlinear programming	Strong adaptability to complex dynamic environments	Difficult to solve and prone to local optima	Solve non-linear task offloading problems in vehicular networks, and handle multi-objective and task scheduling dynamic problems
	Dynamic programming	Precise decision-making and balances computational cost and dynamic adaptability	Computational cost increases with the number of stages and states	Process task dynamic offloading and resource allocation in systems like fog computing and MEC
Heuristic algorithms	Genetic algorithm	Can adapt to dynamic changes and achieve multi-objective balanced optimization	High computational cost and difficult to handle multi-objectives	TODM and perform multi-objective optimization in environments like edge clouds and vehicular networks
	Simulated annealing algorithm	Can escape local optima and find better solutions in dynamic environments	Iterative computational cost increases with the number of tasks and resources	Perform task scheduling and TODM in hierarchical MEC architectures and hybrid cloud-edge systems, and handle single-objective and multi-objective task offloading
	Particle swarm optimization algorithm	Can quickly adapt to changes and improve offloading reliability and efficiency	Computational cost increases with the number of particles and problem scale, and is prone to local optima	Optimize task offloading and resource allocation in scenarios such as dynamic online task offloading, vehicular micro-clouds, and vehicular networks, and handle multi-objective dynamic optimization
Game theory-driven algorithms	Non-cooperative game	Can accurately find optimal offloading and resource allocation strategies	High computational cost, which becomes more prominent when the number of users increases	Solve Nash equilibrium strategies in multi-user task offloading scenarios and analyze resource allocation problems
	Cooperative game	Achieves multi-objective optimization and enhances the ability to cope with dynamic changes	High computational cost and complex algorithm mechanisms	Allocate resources and offloading tasks in multi-agent cooperation scenarios to achieve multi-objective optimization
Machine learning algorithms	Supervised learning	Fast decision-making in stable environments and good at single-objective optimization	The model needs to be frequently rebuilt and adjusted in dynamic environments	Determine offloading strategies based on vehicle dynamic characteristics and solve problems of dynamic network attribute changes
	Unsupervised learning	Can discover potential data patterns and assist in resource allocation and TODM	Optimization problems are prone to gradient disappearance	Reduce the energy consumption of mobile devices in multi-user MEC frameworks and achieve offloading in terrestrial cellular networks
	Reinforcement learning	Can effectively balance and optimize multiple objectives in complex dynamic environments, improving system performance	High computational resource consumption, and the design and adjustment of the reward function affect the algorithm performance	Perform resource management and TODM in scenarios such as HVNet and MEC networks to achieve multi-objective optimization

in the local optima. Dynamic programming can make precise decisions and balance computation and dynamics, but its computational cost increases with scale.

Heuristic algorithms consist of genetic, simulated annealing, and particle swarm optimization algorithms. Genetic algorithms can adapt to dynamic changes and achieve multi-objective optimization in small-scale scenarios; however, their computational cost increases significantly when handling complex multi-objective problems. Simulated annealing algorithms can escape local optima to find better solutions; however, their iterative computational cost is relatively high. Particle swarm optimization algorithms can quickly adapt to changes and improve offloading efficiency, but they are prone to becoming trapped in local optima, and their computational cost also increases.

Game theory-driven algorithms are divided into noncooperative, and cooperative games. Non-cooperative games

can accurately optimize offloading and resource allocation; however, they have a high computational cost, which is more obvious when the number of users increases. Cooperative games can achieve multi-objective optimization and cope with dynamic changes; however, they have high computational costs and complex algorithms.

Machine learning algorithms include supervised, unsupervised, and reinforcement learning. Supervised learning can make quick decisions in stable environments and is good at single-objective optimization; however, it needs to frequently adjust the model in dynamic environments. Unsupervised learning can discover data patterns to assist in decision making, but it is prone to the problem of gradient disappearance during optimization. Reinforcement learning can optimize multiple objectives and improve performance in complex dynamic environments; however, it consumes considerable

computational resources and is affected by the reward function.

To overcome these problems, hybrid algorithms have become a key research direction for meeting the complex requirements of HVNet.

IV. HYBRID ALGORITHMS

In HVNet TODM, the four types of single algorithms, namely traditional, heuristic, game-theory-driven, and machine learning algorithms, each have their own limitations. Therefore, hybrid algorithms have attracted considerable attention in recent years. This section first classifies hybrid algorithms, expounds on various hybrid algorithms and application scenarios, and then analyzes their principles, characteristics, and computational trade-offs. Finally, the advantages and disadvantages of different hybrid algorithms are compared to provide a reference for research and application.

A. TAXONOMY OF HYBRID ALGORITHMS

Hybrid algorithms, which seek to combine the strengths of different algorithms and better adapt to the complex HVNet environment, can be classified into various types based on their core principles and combination methods. Different hybrid algorithms play different roles in TODM and are suitable for different scenarios. Next, we elaborate on the characteristics and applications of various hybrid algorithms.

1) HYBRID OF TRADITIONAL OPTIMIZATION AND MACHINE LEARNING ALGORITHMS

Traditional optimization algorithms can provide precise solutions but lack flexibility. Machine learning algorithms, on the other hand, have strong adaptability. However, in scenarios with limited data or rapid environmental changes, data noise may reduce model accuracy and compromise decision-making stability, i.e., the consistency and reliability of decisions under varying conditions. Combining them integrates precision with adaptability, enabling a collaborative approach to solving TODM problems and creating new avenues for performance improvement. In HVNet task offloading, this hybrid algorithm leverages the strengths of both constituent algorithms, adapting to different task characteristics and network conditions for efficient decision-making.

Cheng's et al. multi-objective dynamic optimization algorithm is particularly effective for complex and dynamic network and task scenarios [111]. In the CEC online task offloading scenario, network traffic scheduling is both non-linear and dynamic, and the multi-objective optimization problem is difficult. Cheng models the problem as a Markov decision process, providing a clear and structured framework for TODM. Simultaneously, an asynchronous deep progressive reinforcement learning method is adopted. The agents in reinforcement learning interact with the environment according to the network state and task dependencies, continuously adjusting offloading and bandwidth decisions. Although the computational cost is large, it successfully solves the problem

of multi-objective dynamic optimization and significantly improves the overall system performance.

In terms of the dynamic changes of network states, Kumar et al. mixes nonlinear programming with Lyapunov-based deep reinforcement learning to adapt to the network changes in HVNet and balance the computational cost and performance [40]. Huang et al. integrates multiple algorithms to address changes in the network state of vehicle edge computing and improve resource utilization [110].

Facing dynamic changes in task characteristics, Pang et al. combines linear programming with supervised learning to adjust decisions quickly according to task characteristics [41]. Wang et al. combines reinforcement learning and dynamic programming to control the computational complexity and adapt to a heterogeneous collaborative edge caching scenario [49].

In the field of dynamic optimization of system costs, Lai et al. uses integer linear programming and a low-complexity heuristic algorithm to balance the computational complexity and cost optimization effects in the MEC scenario [109].

2) HYBRID OF GAME THEORY-DRIVEN ALGORITHMS AND HEURISTIC ALGORITHMS

The hybrid of game theory-driven algorithms and heuristic algorithms aims to merge the analytical power of game theory for modeling competitive/cooperative relationships with the efficient search capability of heuristic algorithms, thereby solving the HVNet TODM problem more effectively. In different scenarios, the collaborative methods of the two have their own characteristics, which can give full play to their respective advantages and improve the efficiency and quality of the TODM.

Sun's et al. work [38] provides a prominent example of algorithm improvement for dynamic resource allocation and task offloading scenarios. When dealing with multi-objective optimization problems, the algorithm integrates a bargaining game, many-to-one matching algorithm, and heuristic iterative calculation process. In the bargaining game link, vehicles and servers flexibly carry out resource allocation and price negotiations according to real-time strategies and utilities, fully considering the interests of both parties and the market dynamics. The many-to-one matching algorithm accurately constructs the matching model between tasks and servers and skillfully incorporates bargaining results to ensure the adaptability of resources to tasks. The heuristic iterative calculation process continuously optimizes the matching relationship through continuous attempts and adjustments, comprehensively considering the dynamic demands of tasks and resources, finding a balance between the computational cost and optimization effect, and effectively improving the efficiency and rationality of resource allocation.

In contrast, Li et al. combines a non-cooperative game and genetic algorithm in the MEC single-objective task offloading scenario to solve problems through the competitive relationship between ES providers (ESP) and device managers

(DM) and the optimization of the genetic algorithm [113]. Wang et al. combines a non-cooperative game, a distributed algorithm (combined with fuzzy logic dynamic pricing), and a genetic algorithm to achieve multi-objective optimization in the multi-objective task offloading scenario of vehicle edge computing [112]. Yuan et al. adopts a hybrid method of game theory analysis, a simulated annealing algorithm, an improved deep Q-network, Markov modeling, and a multi-agent deep deterministic policy gradient algorithm in a multi-user offloading scenario to adapt to the dynamics [55]. Xu et al. combines coalition game theory and the greedy algorithm in D2D traffic offloading to adapt to changes in network topology and resource requirements [58]. Kim mixes a cooperative game model and a particle swarm optimization algorithm for multi-objective task offloading of the UAV-TB-assisted battlefield network platform to achieve multi-agent collaborative competition optimization [11]. Wang et al. mixes a stable matching game algorithm and a dynamic heuristic algorithm in the task offloading scenario to improve the system's social welfare and resource utilization [59].

3) HYBRID OF TRADITIONAL OPTIMIZATION ALGORITHMS AND HEURISTIC ALGORITHMS

The Hybrid of traditional optimization algorithms and heuristic algorithms aims to combine the systematicity and precision of traditional algorithms with the flexibility and fast solution ability of heuristic algorithms. The efficacy of this hybrid approach depends on its integration strategy, which successfully combines the systematic precision of traditional algorithms with the flexible, fast-solving capability of heuristic algorithms. Traditional algorithms are usually based on a rigorous theoretical system and an accurate calculation model, which can provide a stable and accurate basic framework for task offloading and resource allocation. Heuristic algorithms rely on experience and intuitive strategies, and can quickly react and find near-optimal solutions when facing complex and changeable situations. The two cooperate with each other to complement each other's advantages and disadvantages.

In the fields of dynamic resource allocation, task offloading, and multi-objective optimization, the algorithm improvement idea of Kong and Karagiannis is representative [61]. In the TOTFA (Transmission Time Fraction Allocation) scheme, a strategy combining heuristic and traditional optimization algorithms was adopted for the problems of cell association bias and transmission time fraction. The polling algorithm plays an important role in determining cell-association bias. It checks the load status of the cells individually and in order. Once an unbalanced load is found, the incremental adjustment algorithm immediately intervenes and gradually adjusts the cell association according to the real-time load of the base station, thereby effectively balancing the load. When solving the transmission time fraction, the augmented Lagrangian algorithm is used to ensure the stability of service transmission through precise

mathematical operations. This scheme, which mixes the polling, incremental adjustment heuristic algorithm, and augmented Lagrangian algorithm, fully considers the load of the base station and the tolerance of service transmission, finds a balance between load balancing and transmission efficiency, can adapt well to the dynamic changes of the network, and efficiently coordinates resources.

In contrast, Li et al. splits the problem and uses iterative and polling algorithms to achieve resource allocation and task offloading in downlink HCNs based on OFDMA (Orthogonal Frequency-Division Multiple Access) and HVNet scenarios [60]. Ko et al. considers environmental changes in HVNet based on a partially observable Markov decision process model for decision-making trade-offs [63]. Huang et al. uses Lyapunov optimization and a 1-opt local search heuristic algorithm to solve dynamic problems in mobile computing [64]. Song et al. integrates a specific population initialization scheme and an energy-saving scheme in an MEC environment to address the dynamics of HVNet [70]. Misra and Saha uses ILP (Integer linear programming) and a greedy heuristic algorithm to handle multi-hop task offloading in HVNet [71]. Akter et al. combines multiple algorithms in the mGA-TPR algorithm framework to optimize power and resource allocation [86].

4) HYBRID OF GAME THEORY-DRIVEN AND TRADITIONAL OPTIMIZATION ALGORITHMS

The key to hybridizing game theory and traditional optimization algorithms lies in clearly defining the integration boundary. This involves distinguishing between the role of game theory in modeling multi-agent strategic interactions and the role of traditional optimization in finding efficient solutions. This hybrid method uses game theory to construct a model of the competitive and cooperative relationships among multiple agents, and uses theories such as stable matching and coalition formation to ensure the stability of decisions and individual rationality. At the same time, it combines traditional algorithms such as integer programming and greedy strategies to solve the optimal solution of resource allocation within the game framework, and introduces a dynamic adjustment mechanism to balance global search and local optimization through strategies such as random initialization and iterative exchange.

In research on the dynamic perception scheduling of resources and tasks, the bee colony-based task offloading in the vehicular edge computing (BTV) algorithm of de Souza et al. is unique [62]. In the HVNet environment, information on resources and tasks changes rapidly. The BTV algorithm is based on an Artificial Bee Colony (ABC) meta-heuristic algorithm. The ABC algorithm simulates the foraging behavior of bees: the leading bee's search for food sources is analogous to searching for resources, while the follower bees' selection of food sources is analogous to allocating resources. This mechanism allows the algorithm to accurately perceive the complex and change-

able resource and task situations. At the same time, game theory, a powerful tool, is introduced to analyze the strategy spaces and payoff functions of various agents to coordinate the interests of various agents in HVNet and fully ensure the efficient utilization of resources and the reasonable offloading of tasks. However, in the scenario of large-scale tasks, the computational cost of the ABC algorithm increases explosively, and the coordination difficulty of game theory also increases, resulting in a significant extension of the decision-making time. Once the network topology changes frequently, the adaptability of the algorithm is inadequate, and the accuracy of resource allocation is greatly reduced.

In contrast, Liu et al. distinguished between indivisible and divisible tasks according to the changes in the characteristics of tasks in HVNet and MEC and adopts different algorithms for decision-making to balance task performance and energy consumption, but did not integrate game theory and traditional algorithms [65]. Adhikari et al. uses the accelerated particle swarm optimization (APSO) algorithm based on particle swarm optimization to search for the optimal solution in the dynamic environment of HVNet, but did not clearly show the hybrid of game theory and traditional algorithms, and is also prone to fall into a local optimum [85]. Mahjoubi et al. combines a simulated annealing algorithm and a prediction algorithm for dynamic task scheduling, did not use game theory, and had defects in scenarios of rapid changes in network topology and large-scale tasks [23].

5) HYBRID OF MACHINE LEARNING ALGORITHMS AND HEURISTIC ALGORITHMS

While machine learning algorithms offer powerful learning and adaptation, heuristic algorithms are highly efficient due to their reliance on heuristic rules and local search. The combination of the two aims to deal more effectively with the complex and changeable environment of HVNet and solve the TODM problem.

Wang's et al. research results are outstanding for dealing with various dynamic change scenarios. In the task scheduling of the pervasive edge computing network, the multi-agent imitation learning-based computing offloading algorithm (MILP) algorithm integrates multiple technologies [67]. Convolutional neural networks (CNN) and generative adversarial networks (GAN) first process tasks, and environmental information accurately extracts key features from a large amount of historical data. Based on this, stochastic game theory and reinforcement learning effectively predict the changes in task demands and device states through the interaction between agents and the environment. Then, heuristic algorithms such as the genetic algorithm encode the task allocation scheme as a chromosome according to the prediction and use selection, crossover, and mutation operations to fully search for better strategies in the solution space, aiming to minimize the average task completion time of devices while considering task urgency.

In contrast, Luo et al., in the dynamic decision-making scenario of vehicles in HVNet, with the help of the self-learning-based distributed computing offloading (DISCO) scheme, based on the game theory model, analyzes data through machine learning to predict the changes in vehicle decisions, and then uses the particle swarm optimization algorithm to adjust TODM [69]. Cui et al. for the dynamic changes in the multi-user network environment, constructs the task offloading problem as a game model, uses machine learning for prediction, and optimizes the task offloading strategy with algorithms such as the genetic algorithm [78]. Xie et al., in the scenario of dynamic update of network node strategies, combines game theory and learning algorithms to predict the changes in node strategies through machine learning, and the particle swarm optimization algorithm adjusts the search direction of node strategies accordingly [88].

6) HYBRID OF GAME THEORY-DRIVEN AND MACHINE LEARNING ALGORITHMS

The core of the hybrid game theory and machine learning algorithms is the construction of a multi-agent system and a strategic interaction framework. Game theory is used to analyze the payoff situations of each decision-making subject under different strategy combinations, and machine learning algorithms are used to learn from historical data and environmental feedback to predict the optimal strategy and adjust the decision-making strategy. Thus, the task offloading strategy can be continuously optimized in a complex environment to achieve reasonable resource allocation and efficient task offloading.

In the research field of task scheduling dynamics, Wang's et al. MILP algorithm stands out [67]. In complex task scheduling scenarios, such as HVNet, this algorithm integrates a variety of cutting-edge technologies. Stochastic game theory provides a basic framework for the interaction between agents, and agents, such as vehicles and ES, conduct strategic interactions within this framework. CNN and GAN are responsible for processing a large amount of task and environment information, and extracting key features from it. Generalized adversarial imitation learning (GAIL) focuses on learning expert experience and provides a better decision-making reference for algorithms. ACKTR (actor-critic with Kronecker-factored trust region) optimizes the strategy update to ensure the efficiency and stability of the learning process. Through continuous learning, the machine learning algorithm continuously optimizes the task scheduling strategy and feeds the optimization results back to the stochastic game theory to further adjust the payoff function and strategic interaction. Although this process significantly increases the computational cost, it can accurately adapt to the dynamic changes in task scheduling and effectively minimize the average task completion time of devices.

In contrast, Luo's et al. DISCO scheme focuses on HVNet TODM, combines game theory and machine learning, and uses relevant mechanisms to address the dynamics of HVNet,

minimizing the joint cost of vehicles with moderate computational complexity [69]. Cui et al. uses evolutionary game theory and reinforcement learning in the dynamic optimization of multi-user offloading strategies, and meets the optimization requirements through their interaction; however, reinforcement learning training increases the computational cost [78]. The MBR-CO (Myopic Best Response-Based Computing Offloading) and SL-CO (Stochastic Learning-Based Computation Offloading) algorithms proposed by Xie et al. combine potential games and learning algorithms, and through specific strategy update methods, achieve the dynamic optimization of multi-user offloading strategies with relatively flexible calculations, balancing dynamic adaptability and computational efficiency [88].

7) HYBRID OF ALGORITHMS OF THE SAME TYPE

A hybrid of algorithms of the same type means that in the HVNet TODM algorithm, different algorithms of the same category are combined. This hybrid method can more effectively solve complex problems and improve the algorithm performance and dynamic adaptability by integrating the different advantages of algorithms of the same type.

When dealing with various dynamic change scenarios, Li's et al. algorithm improvement concept is very representative [68]. In the complex dynamic environment of HVNet, Li skillfully combines the Markov decision process (MDP), the deep deterministic policy gradient (DDPG) algorithm, the actor-critic architecture, and the policy gradient algorithm. The MDP accurately models the dynamic environment of the HVNet, clearly presenting the complex network state. The DDPG algorithm finely adjusts power scheduling according to the real-time state of the device. Between the contradiction of energy consumption and task execution efficiency, the DDPG ensures that task execution is both efficient and stable through continuous trade-offs. For example, when the device's battery level is low, the DDPG appropriately reduces power consumption and gives priority to ensuring the completion of the task; when the network condition is good and the device performance is sufficient, it can increase the power and speed up task execution.

In contrast, Ali et al. integrates the encoder-decoder, MLP, and LSTM to process the characteristics of dynamic information, capture the change trend, adapt to dynamic changes such as network bandwidth, and balance energy saving and task execution [74]. Vemireddy and Rout [79] combines reinforcement learning and fuzzy logic. The former makes the RL agent act according to the state to obtain rewards, and the latter calculates the weight according to the dynamic parameters of the vehicle and adjusts the task allocation strategy to deal with changes in the vehicle state [79]. Bi et al. uses the GSP algorithm to jointly optimize multiple parameters and balance energy consumption and network latency [25]. Liao et al. mixes the DDPG power scheduling and the Double DQN task offloading algorithm, and according to the dynamics of the device, network, and tasks, trades off power consumption

and task execution time [66]. You et al. mixes the many-to-many matching theory and the multi-round sealed sequential auction mechanism to achieve reasonable resource allocation and TODM [92]. Li et al. combines deep reinforcement learning, the attention mechanism, and the communication channel, and adjusts the strategy according to the network state to optimize task offloading and resource utilization [93]. Dong et al. mixes particle swarm optimization (PSO) and quantum particle swarm optimization (QPSO) are used to find a low-energy-consumption task offloading scheme and improve the dynamic performance of the algorithm [26]. Shu and Li improves QPSO and introduces a crossover operator and an adaptive coefficient to adapt to the dynamic changes in task offloading [28]. Chen et al. mixes QPSO and NSGA-II (Non-dominated Sorting Genetic Algorithm II) to enhance the algorithm's ability to deal with dynamic scenarios [54]. Yuan et al. combines the simulated annealing criterion and migratory bird optimization algorithm to improve the dynamic search ability [24].

8) HYBRID OF THREE OR MORE HETEROGENEOUS ALGORITHMS

A hybrid of multiple heterogeneous algorithms combines different types of algorithms—such as traditional, heuristic, machine learning, and game-theoretic approaches—within HVNet TODM. This hybrid method integrates the advantages of multiple algorithms, performs well for different performance indicators, better solves the TODM problem, and adapts to the complex environment of HVNet.

In dealing with the complex challenges of a dynamic environment, Wang's et al. algorithm improvement idea is particularly outstanding. In a task offloading scenario, Wang organically integrates multiple algorithms [87]. The MDP theory clearly defines the structure of the task offloading problem, providing a clear framework for the complex task process. The mean field game (MFG) focuses on handling the dynamic strategic interactions among multiple agents, accurately analyzing the strategic changes of each agent and their mutual influences in a complex environment with multiple agents participating. The GAN algorithm fully exploits its unique advantages, generates a reasonable task allocation ratio, and trains the value function, making the task allocation more scientific and efficient. The water-filling algorithm skillfully allocates the rate according to the characteristics of the task. For example, for urgent tasks, a higher rate is preferentially allocated to ensure rapid completion, whereas for general tasks, a balance is sought between resource utilization and computational efficiency. Through the coordinated operation of these algorithms, a delicate trade-off is achieved among key factors, such as task urgency, resource utilization, and computational efficiency, and the optimal task offloading scheme is successfully achieved.

In contrast, He et al. combines methods such as queuing theory to describe task processing and waiting times, and uses relevant conditions to solve optimization problems, balancing

task completion time, and resource utilization [89]. Song et al. mixes multiple algorithms to dynamically tradeoff between bandwidth allocation and task offloading latency [90]. Cardellini et al. combines game theory and queuing theory, integrates multiple technologies, and dynamically trades off resource competition, task execution efficiency, and distributed computing costs [94]. Mao et al. combines game theory with the HHGA-SA hybrid algorithm to find the optimal solution while satisfying dynamic constraints and balancing costs in multiple aspects [45].

B. ANALYSIS OF HYBRID ALGORITHMS

Owing to their unique advantages, hybrid algorithms have emerged as a research focus in HVNet TODM. They combine the strengths of multiple algorithms to effectively address the complexities of the HVNet environment. This section analyzes the innovative principles and computational trade-offs of the new framework of hybrid algorithms and conducts a comprehensive comparison of various hybrid algorithms.

1) PRINCIPLE ANALYSIS

a: ALGORITHM HYBRID DESIGN AND COMPLEMENTARY ADVANTAGES

Hybrid algorithms leverage the strengths of different algorithms to address the challenging problem of task offloading in HVNet. The hybrid algorithm taxonomy framework proposed in this paper differs fundamentally from existing frameworks (e.g., [57]).

Reference [57] focuses on task offloading in fog computing, and its taxonomy revolves around TODM factors, optimization techniques, and common algorithm categories. Its taxonomy of hybrid algorithms is not sufficiently detailed, and it does not fully consider the dynamic coordination of algorithms and cross-layer optimization of HVNet.

This framework begins with the core principles of algorithms and divides them into five major categories. The subdivision of the hybrid algorithms includes various complex types. For example, a hybrid of traditional optimization and machine learning algorithms. Traditional optimization algorithms (such as linear programming) can construct accurate task offloading schemes; however, it is difficult to adapt to the dynamic changes in HVNet. Machine learning algorithms (such as reinforcement learning) are highly adaptable. When the two are combined, the traditional algorithm lays the decision-making framework, and the machine learning algorithm adjusts the strategy according to real-time data. These features are critical in autonomous driving scenarios, where the traditional algorithm constructs a basic offloading framework based on map information and traffic rules, and the machine learning algorithm uses real-time sensor data to dynamically adjust the strategy. This type of close and efficient dynamic coordination is difficult to achieve in the framework of reference [57].

The hybrid of game theory and heuristic algorithms is also a feature of this framework. Game theory-driven algorithms

can optimize resource allocation; however, the calculation is complex in large-scale scenarios. Heuristic algorithms can quickly determine near-optimal solutions. When the two collaborate, game theory constructs the model and heuristic algorithms search, thereby improving the efficiency of resource allocation. The existing framework may not be able to fully exploit the advantages of the two and achieve similar efficient coordination in this regard.

b: THE IMPACT OF HYBRIDIZATION ON ALGORITHM CHARACTERISTICS

Hybridization has a significant impact on the convergence, stability, and computational complexity of algorithms, and there are differences between the new and existing frameworks in this regard.

Traditional algorithms converge slowly in large-scale scenarios, heuristic algorithms are prone to falling into local optima, and the convergence of machine learning is affected when dealing with big data. The hybrid algorithms under the new framework performs the best. For example, in a hybrid of traditional optimization and machine learning, they collaborate to converge quickly in small-scale scenarios, and in large-scale scenarios, machine learning helps traditional algorithms optimize the search direction. In a hybrid of game theory and heuristic algorithms, heuristic algorithms accelerate convergence, and game theory improves convergence quality. Because the existing framework does not fully consider the deep integration and coordination among algorithms, when dealing with large-scale and complex scenarios, its optimization effect on the algorithm characteristics is not as obvious as that of the new framework.

In terms of stability, it is difficult for traditional algorithms to adapt to the dynamics of HVNet, heuristic algorithms perform poorly in complex scenarios, game theory is likely to cause instability in large-scale scenarios, and machine learning is restricted by data quality. The new framework enhances stability through integration. For example, in a hybrid of traditional optimization and machine learning, machine learning adjusts the decisions of traditional algorithms according to the environment. In a hybrid of game theory and heuristic algorithms, game theory reduces conflicts, and heuristic algorithms quickly respond to environmental changes. By contrast, when dealing with the stability problem of algorithms, the existing framework lacks a comprehensive and effective integration strategy similar to that of the new framework.

In terms of computational complexity, the computational complexity of multiple algorithms is relatively high in large-scale scenarios. Hybrid algorithms under the new framework are optimized through the division of labor. When traditional optimization and machine learning are combined, traditional algorithms handle the regular part, and machine learning provides parameters and strategies. In the hybrid of game theory and heuristic algorithms, heuristic algorithms reduce the computational cost of game theory, creating the possibility of the application of HVNet devices. In terms of optimizing the computational complexity, the existing framework may not

fully tap the collaborative potential among algorithms, and it is difficult to achieve the optimization effect of the new framework.

2) COMPARISON OF HYBRID ALGORITHMS

In the research on HVNet TODM algorithms, hybrid algorithms have prominent advantages but also face challenges. Different types of hybrid algorithms have advantages and disadvantages in practical scenarios. The following makes a detailed comparison of various hybrid algorithms from aspects such as algorithm principles and integration methods, performance, and applicable scenarios, as Table 3.

a: HYBRID OF TRADITIONAL OPTIMIZATION AND MACHINE LEARNING ALGORITHMS

This type of hybrid algorithm combines the precision of traditional optimization algorithms and the adaptability of machine learning algorithms. Traditional optimization algorithms (such as linear programming and nonlinear programming, etc.) can build a relatively accurate framework for task offloading schemes based on precise mathematical models, providing a stable basis for decision-making. Machine learning algorithms (such as reinforcement and supervised learning, etc.) can automatically adjust strategies according to real-time data to adapt to the complex and changeable environment of HVNet.

In practical applications, we consider an autonomous driving scenario as an example. The traditional algorithm builds a basic framework to determine the rules and directions of task offloading. The machine learning algorithm then dynamically adjusts the offloading strategy according to the data collected in real time using sensors, such as the vehicle location, speed, and surrounding environment information. In this way, it not only ensures a certain degree of decision-making accuracy, but also improves the system's adaptability to the dynamic environment, thereby increasing resource utilization and task completion rate. However, this hybrid approach faces challenges. Machine learning algorithms rely heavily on high-quality data. Noisy or insufficient data can impair decision-making accuracy, leading to poor coordination and adaptation between the constituent algorithms. In the future, it can be improved by optimizing the data processing flow, designing dynamic weight strategies, etc., and has great development potential in scenarios such as intelligent driving and vehicle-road coordination.

b: HYBRID OF GAME THEORY-DRIVEN ALGORITHMS AND HEURISTIC ALGORITHMS

The hybrid of game theory-driven algorithms and heuristic algorithms aims to combine the game theory's advantage in analyzing competitive and cooperative relationships with heuristic algorithms' fast search ability. game theory-driven algorithms analyze the competitive and cooperative relationships among participants, such as vehicles, ES, and cloud servers, and construct a reasonable resource allocation model to ensure the stability of decisions and individual rationality.

Heuristic algorithms, relying on heuristic rules and local search strategies, can find near-optimal solutions within a short time frame, thereby improving search efficiency.

In intelligent logistics, game theory-driven algorithms can model the objectives and requirements of various entities and allocate transportation resources accordingly. Heuristic algorithms then quickly plan the transportation route according to real-time traffic conditions, vehicle loads, and other information to improve logistics efficiency. However, in large-scale scenarios, the computational complexity of game theory-driven algorithms increases significantly, and heuristic algorithms are likely to fall into local optimal solutions. In the future, distributed computing technology can be introduced and the heuristic search strategy can be improved, which will play an important role in the field of intelligent logistics and others.

c: HYBRID OF TRADITIONAL OPTIMIZATION ALGORITHMS AND HEURISTIC ALGORITHMS

The Hybrid of traditional optimization and heuristic algorithms integrates the systematic nature and precision of traditional algorithms with the flexibility and fast solution ability of heuristic algorithms. Traditional algorithms, based on a rigorous theoretical system and an accurate calculation model, provide a stable and accurate basic framework for task offloading and resource allocation. Heuristic algorithms, which rely on experience and intuitive strategies, can quickly react and find near-optimal solutions when facing complex and changeable situations.

In the dynamic resource allocation scenario of HVNet, such as in emergency communication, traditional algorithms can be used to build a communication framework and determine the basic rules of resource allocation. Heuristic algorithms, according to the urgency and network conditions, quickly select the communication path and allocate resources to ensure timely information transmission. However, in a complex network environment, the decision-making accuracy and stability of this hybrid algorithm must be improved. In the future, machine learning technology can be used to optimize heuristic rules and enhance the dynamic adjustment ability of traditional algorithms, which have a high application value in fields such as intelligent transportation and edge computing.

d: HYBRID OF GAME THEORY-DRIVEN AND TRADITIONAL OPTIMIZATION ALGORITHMS

The key to the hybrid of game theory and traditional optimization algorithms lies in accurately defining the integration boundary between game theory modeling the strategic interaction of multiple agents and the efficient solution of traditional optimization algorithms. Game theory is used to construct a model of competitive and cooperative relationships among multiple agents, and theories such as stable matching and coalition formation are used to ensure the stability of decisions and individual rationality. Traditional optimization algorithms (such as integer programming and

TABLE 3. Comparative analysis of hybrid task offloading decision-making algorithms.

Types of Hybrid Algorithms	Advantages	Disadvantages	Applicable Scenarios
Hybrid of traditional optimization and machine learning Algorithms	High decision-making accuracy, strong adaptability, and improved resource utilization and task completion rate.	Highly dependent on high-quality data. Poor coordination and adaptability when data quality is poor.	Autonomous driving and vehicle-road coordination scenarios.
Hybrid of game theory and heuristic algorithms	Strong ability to handle interest relationships, high search efficiency, and reasonable resource allocation.	High computational complexity in large-scale scenarios. Heuristic algorithms are prone to getting trapped in local optima.	Intelligent logistics field.
Hybrid of traditional optimization algorithms and heuristic algorithms	Both systematic nature and flexibility, and can balance load and transmission efficiency.	Decision-making accuracy and stability need to be improved in complex networks.	Dynamic resource allocation scenarios in intelligent transportation and edge computing.
Hybrid of game theory and traditional optimization algorithms	Reasonable resource allocation in multi-user scenarios.	Difficult coordination in game theory and high computational complexity of traditional algorithms lead to decision-making delays in large-scale tasks.	The integration scenario of HVNet and the industrial Internet of Things.
Hybrid of machine learning and heuristic algorithms	Strong ability to cope with complex environments and improve traffic efficiency in multi-vehicle interactions.	Long training time for machine learning algorithms. Heuristic algorithms lack a global optimal solution.	Autonomous driving and traffic management scenarios.
Hybrid of game theory and machine Learning algorithms	Continuously optimize task offloading strategies in complex environments and coordinate the interests of all parties.	Insufficient lightweighting, heavy burden on edge devices, and low coordination efficiency.	Intelligent logistics and multi-vehicle collaborative operation scenarios.
Hybrid of algorithms of the same type	Solve complex problems and improve algorithm performance and dynamic adaptability.	Complex combinations increase computational complexity, and simple combinations have limited adaptability.	Task scheduling scenarios such as hybrid road networks in logistics and autonomous driving.
Hybrid of multiple heterogeneous algorithms	Good performance in multiple indicators and can optimize urban traffic, etc.	Insufficient lightweighting and integration with edge computing.	Complex scenarios such as urban traffic and the collaboration of intelligent connected vehicles.

greedy strategy, etc.) solve the optimal solution of resource allocation within the game framework and balance the global search and local optimization by introducing a dynamic adjustment mechanism.

In the multiuser HVNet scenario, the hybrid algorithm can allocate resources reasonably. However, in large-scale task scenarios, it is difficult to coordinate game theory and traditional algorithms are computationally complex, resulting in decision-making delays. In the future, by simplifying the game theory model and optimizing the solution process of traditional algorithms, better results can be achieved with the integration of HVNet and the Industrial Internet of Things.

e: HYBRID OF MACHINE LEARNING ALGORITHMS AND HEURISTIC ALGORITHMS

Machine learning algorithms have powerful learning and adaptive capabilities, and heuristic algorithms exhibit high efficiency, relying on heuristic rules and local search. The combination of the two aims to deal more effectively with the complex and changeable environment of the HVNet. Machine learning algorithms mine potential patterns and laws in the data by learning from a large amount of historical data and predicting task demands and environmental changes. Heuristic algorithms then make decisions quickly, according to real-time situations.

In the multivehicle interaction scenario of intelligent transportation, machine learning algorithms learn traffic patterns, and heuristic algorithms make decisions according to

real-time traffic conditions to improve traffic efficiency at intersections. However, machine learning algorithms require a long training time, and heuristic algorithms lack a global optimal solution. In the future, online learning technology can be studied and a global optimization strategy can be introduced to improve heuristic algorithms, which has obvious advantages in fields such as autonomous driving and traffic management.

f: HYBRID OF GAME THEORY-DRIVEN AND MACHINE LEARNING ALGORITHMS

The core of the hybrid game theory and machine learning algorithms is the construction of a multi-agent system and a strategic interaction framework. Game theory analyzes the payoff situations of each decision-making subject under different strategy combinations, and machine learning algorithms learn from historical data and environmental feedback to predict the optimal strategy and adjust the decision-making strategy; thus, the task offloading strategy can be continuously optimized in a complex environment.

In the logistics path optimization scenario, game theory coordinates the interests of all parties, and machine learning plans the path according to traffic conditions and cargo demands. Currently, this algorithm has problems, such as insufficient lightweighting, heavy burden on edge devices, and low coordination efficiency. In the future, model compression technology can be adopted, and the coordination mechanism can be optimized, which will result in excellent

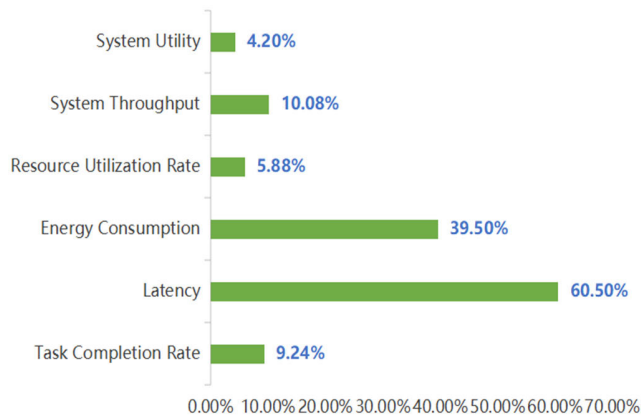


FIGURE 4. Statistical distribution of performance indicators coverage in HVNet TODM literature.

performance in scenarios such as intelligent logistics and multi-vehicle collaborative operations.

g: HYBRID OF ALGORITHMS OF THE SAME TYPE

A hybrid of algorithms of the same type combines different algorithms of the same category and integrates their advantages. For example, in a multi-ES task scheduling scenario, different reinforcement learning algorithms or intelligent optimization algorithms work together to allocate tasks according to server load and task priority, improving the processing efficiency.

However, the combination of complex algorithms leads to an increase in the computational complexity, and the adaptability of simple combinations is limited. In the future, an efficient coordination mechanism needs to be explored to balance the computational complexity and adaptability in scenarios such as a hybrid road network of logistics and autonomous driving.

h: HYBRID OF THREE OR MORE HETEROGENEOUS ALGORITHMS

The hybrid of multiple heterogeneous algorithms integrates the advantages of multiple algorithms and can perform well for different performance indicators. Consider urban traffic governance as an example: The integration of the Markov decision process, mean field game, generative adversarial network, and water-filling algorithm enables the coordinated regulation of traffic flow, optimize traffic lights and vehicle routes, and relieve congestion.

Currently, there are problems, such as insufficient lightweighting and edge computing integration. In the future, the algorithm structure can be optimized, and edge computing optimization technology can be adopted to promote the development of intelligent transportation in scenarios such as urban traffic and collaboration of intelligent connected vehicles.

In summary, as shown in Table 3, the hybrid algorithms in the TODM of HVNet can integrate the advantages of different

algorithms to improve the performance, yet they also have limitations.

The combination of traditional optimization and machine learning algorithms can enhance resource utilization and task completion rates in scenarios such as autonomous driving. However, it relies on high-quality data, and when the data quality is poor, synergy is poor. The combination of game theory and heuristic algorithms can reasonably allocate resources to intelligent logistics. However, in large-scale scenarios, the calculation is complex, and heuristic algorithms are prone to becoming trapped in local optima. The combination of traditional and heuristic algorithms can balance load and transmission efficiency in scenarios such as intelligent transportation. However, in complex networks, decision-making accuracy and stability must be improved. The combination of game theory and traditional optimization algorithms is applicable to the integration of vehicular networks and industrial IoT. In large-scale tasks, it is difficult to coordinate, and there is a decision-making delay. The combination of machine learning and heuristic algorithms can improve the traffic efficiency in scenarios such as autonomous driving. However, machine learning requires a long training time, and heuristic algorithms lack global optimal solutions. The combination of game theory and machine learning algorithms can optimize offloading strategies in scenarios such as intelligent logistics; however, there are problems such as insufficient lighting. A combination of algorithms of the same type can solve complex problems in task-scheduling scenarios, such as logistics. Complex combinations increase the computational complexity, and simple combinations have limited adaptability. The combination of multiple heterogeneous algorithms can perform well for multiple indicators in complex scenarios such as urban traffic, but there are problems such as insufficient lightweighting and edge computing integration.

Given the various problems that hybrid algorithms currently face in the TODM of HVNet, the next section conducts an in-depth exploration of targeted optimization strategies and future development trends, striving to provide new ideas for promoting technological progress in this field.

V. ANALYSIS OF RESEARCH TRENDS

This section mainly discusses the literature coverage of such dimensions as performance indicators and key characteristics from the perspective of coverage, and analyzes the research trends.

A. PERFORMANCE INDICATORS

The experiments in the surveyed literature primarily focus on task offloading, employing multiple performance indicators to evaluate the performance of algorithms and schemes. Fig. 4 statistically analyzes the usage of typical performance indicators based on the proportion of studies.

Analysis shows that research attention varies significantly across different performance indicators. Task execution latency and energy consumption are the primary focus

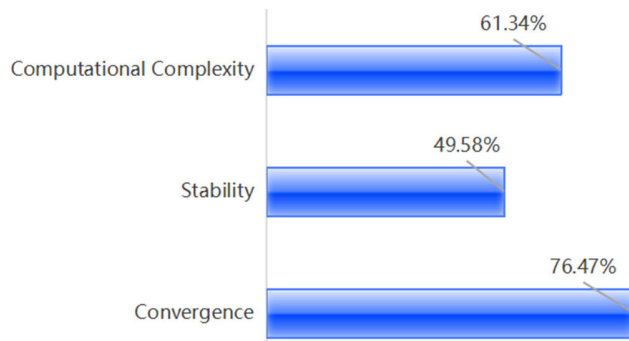


FIGURE 5. Analysis of key algorithm characteristics coverage in HVNet TODM research.

areas, accounting for 60.50% and 39.50% of relevant papers, respectively. For instance, vehicle tasks in [109] require quick responses and low energy consumption, these two indicators have become key objects of attention. The task completion rate and system throughput are also relatively high, accounting for 9.24% and 10.08%, respectively. These reflect the processing capabilities of the system from different perspectives. As mentioned in [35], high values of these indicators are helpful for improving network performance.

In contrast, indicators such as resource utilization, system utility, and fairness receive less attention, with proportions of 5.88%, 4.20%, and 3.36%, respectively. With the deepening of research, overall system performance and fairness are gradually receiving increasing attention [82].

The limited computing resources in HVNet explain why current research primarily focuses on task processing efficiency and resource optimization. Indicators such as the task completion rate and system throughput reflect the attention paid to the improvement in the overall system performance. As the complexity of application scenarios increases, comprehensive evaluation with multiple indicators has become a trend, and some studies have already adopted this approach to evaluate algorithm performance [111].

The current research landscape presents several challenges. Owing to the lack of attention to indicators such as resource utilization and fairness, the research depth may not be sufficient. In practical applications, reasonable utilization and fair distribution of resources are very important for the stable development of the system, and more research is needed in this regard. The standardization of indicators also needs to be strengthened. Different studies may have different definitions and calculation methods for the same indicator, which is not conducive to the comparison and promotion of research. Thus, a unified standard must be established.

B. KEY CHARACTERISTICS

In the complex and dynamically changing network scenarios of HVNet, the computing capabilities, communication conditions, and task requirements of the different types of vehicles vary significantly. This results in higher requirements for the key characteristics of algorithms, such as convergence,

stability, and computational complexity. The following section presents an in-depth analysis combined with the characteristics of HVNet. Fig. 5 statistically analyzes the coverage of the key characteristics.

In HVNet TODM research, convergence is the most studied characteristic, covered in 76.47% of the papers. Convergence is a core indicator of the algorithm performance and is related to the accuracy and effectiveness of TODM. In an actual environment with diverse tasks, dynamically changing resources, and a complex network, the requirements for the convergence ability of the algorithms are extremely high. Traditional algorithms exhibit significant differences in convergence performance in scenarios of different scales. In small-scale scenarios, good results can be obtained through a limited number of iterations, but in large-scale and complex scenarios, the computational cost increases sharply, and the convergence speed slows. Heuristic algorithms can quickly find approximate solutions in small-scale scenarios, but they are prone to falling into local optima in large-scale scenarios, and the convergence quality decreases. The convergence speed and stability of machine learning algorithms under different network conditions and data environments have been widely studied, and the relevant achievements provide theoretical support for algorithm optimization and application.

Research on stability account for only 49.58% of the total, which is relatively low. This indicates that there is still much room for improvement in the research on how algorithms cope with system changes and external interference. In HVNet, stability is a key guarantee for the reliable operation of algorithms, and there are many factors related to system changes and external interferences. Some papers have discussed the performance of algorithms in complex situations. For example, the SATS algorithm in [23] shows good stability, but some simple rule-based algorithms exhibit poor stability in complex environments. The stability of game-theory-driven algorithms is affected in large-scale scenarios, and machine learning algorithms rely strongly on data quality and model adaptability; thus, stability also faces challenges. Therefore, it is crucial to strengthen the research on algorithm stability.

Research on computational complexity accounts for 61.34%, which is at an intermediate level. This indicates that relevant research has a wide focus on the computational resources required for algorithm execution. Computational complexity is an important indicator for measuring algorithm efficiency and affects the feasibility and scalability of practical applications of algorithms. Different types of algorithms exhibit different performance in terms of time and space complexity. There are obvious differences in the time complexity between simple algorithms and deep reinforcement learning algorithms, and there are also differences in the space complexity between algorithms based on the greedy strategy and some algorithms based on neural networks. A comparative study of the time complexity of some algorithms in [109] provides a reference for algorithm selection, but further exploration is still needed to meet the requirements of

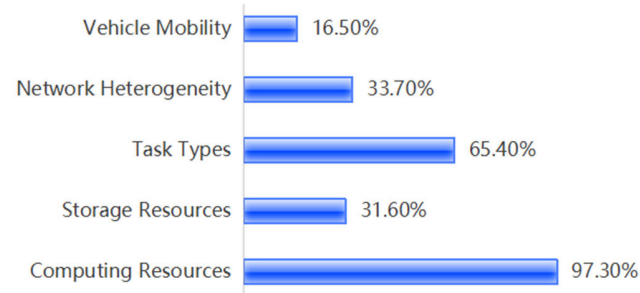


FIGURE 6. Coverage statistics of key influencing factors in HVNet TODM studies.

the efficient utilization of algorithm computational resources in complex application scenarios.

C. KEY FACTORS

Existing studies have shown significant differences in the coverage of key factors in HVNet TODM algorithms (Fig. 6). Analysis of the correlations between them is generally insufficient (Table 3). This reflects the current situation of the consideration of key factors in the current research and points out directions for subsequent research.

1) SIGNIFICANT DIFFERENCES IN THE COVERAGE OF KEY FACTORS

The coverage of computing resources is as high as 97.3% in existing research, which can be attributed to their direct impact on task-processing capabilities. In TODM, the allocation and utilization of computing resources play a decisive role in the algorithm performance; therefore, they have received extensive attention from researchers.

By contrast, the coverage of storage resources and network heterogeneity is only approximately 30%, and the coverage of vehicle mobility is even as low as 16.5%. Storage resources are of great significance in task offloading; however, the research focus is mostly on computing and communication aspects, resulting in insufficient attention. The environments of network heterogeneity and vehicle mobility are complex and changeable, and the research difficulty is high, making their coverage in the existing research relatively low.

The coverage of task types is 65.4%, occupying an important position in TODM. Different types of tasks have different requirements for computing, storage, and network resources, which affect the formulation of TODM. However, current research on task types is not sufficiently in-depth and comprehensive, and the specific influence mechanism of task types on algorithms in different scenarios has not been fully explored. The imbalance of attention to key factors in current research will make it difficult for the algorithm design to fully consider the needs of actual scenarios.

2) GENERAL LACK OF CORRELATION ANALYSIS

The degree of lack of correlation analysis among the various key factors is relatively high, as shown in Table 4. In fact,

TABLE 4. Degree of insufficient correlation analysis among key factors in HVNet TODM research.

Key Factors	Lack of Correlation Analysis
Computing Resources	High
Storage Resources	High
Task Types	High
Network Heterogeneity	High
Vehicle Mobility	High

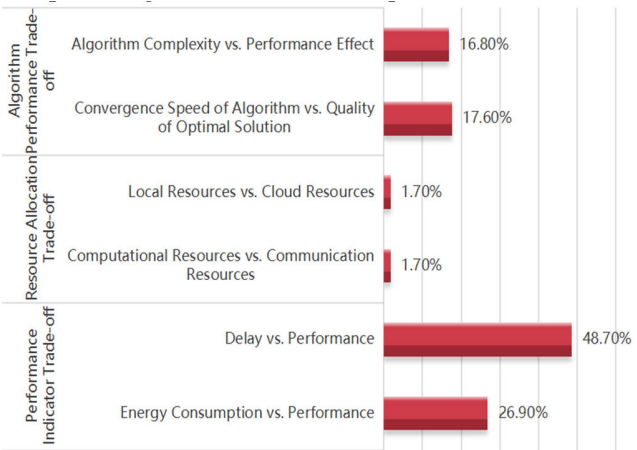


FIGURE 7. Distribution of research coverage on computational trade-offs in HVNet TODM.

the computing resources are closely related to other factors. Network heterogeneity affects the transmission efficiency of computing resources, and vehicle mobility changes the allocation requirements of computing resources. The task type is also closely related to other factors. Different task types determine different requirements for computing and storage resources, and at the same time, they will also be affected by changes in network heterogeneity and vehicle mobility.

For example, in environments with high network heterogeneity and high vehicle mobility, real-time interactive tasks may suffer from increased data transmission latency due to network instability, compromising their real-time requirements. Currently, a more reasonable selection of offloading strategies and resource-allocation methods is required. However, current research does not pay enough attention to this issue. This is mainly because it is relatively simple to study a single factor alone, whereas comprehensively considering the correlations among multiple factors requires constructing more complex models and conducting more difficult experiments. However, in an actual HVNet environment, these factors influence and interact with one another. If the correlations between them are ignored, the performance of the algorithms in practical applications is seriously limited.

D. COMPUTATIONAL TRADE-OFFS

Computational trade-offs are crucial in HVNet TODM and permeate key dimensions such as performance indicators, resource allocation, and algorithm performance. Different

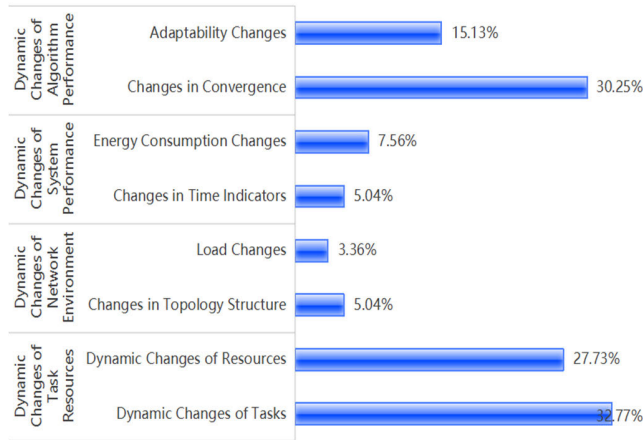


FIGURE 8. Research attention on dynamic characteristics in HVNet TODM algorithms.

types of algorithms have their own characteristics when dealing with these tradeoffs.

The experiments in the cited papers show that the problem of computational trade-offs runs through the entire research process and plays a decisive role in the overall performance of the system. Fig. 7 shows the proportion of papers for each trade-off type, which helps grasp the research focus and trends and provides guidance for relevant explorations.

In terms of performance indicators, the trade-offs between energy consumption and performance as well as between latency and performance have become the core focus of this research. The proportions of relevant papers are 26.9% and 48.7%, respectively, and the sum of the two is as high as 75.6%, highlighting their crucial position.

In terms of resource allocation, the proportion of papers on the trade-offs between computing and communication resources, as well as between local and cloud resources, is only 1.7%, with a total of 3.4%. The degree of attention is significantly lower than that of the performance indicator trade-offs, but their practical significance cannot be underestimated.

In terms of algorithm performance, the proportions of papers on the trade-offs between the convergence speed of the algorithm and the quality of the optimal solution, as well as between the algorithm complexity and the performance effect, are 17.6% and 16.8%, respectively, and the sum of the two is 34.4%. This indicates that researchers attach great importance to the balance of the algorithm performance during the process of algorithm design and optimization.

E. DYNAMIC CHARACTERISTICS

The dynamic characteristics cover four core dimensions: task resources, network environment, system performance, and algorithm performance. These dimensions are interrelated and jointly form a complex ecosystem for system operations.

The dynamics of task resources are reflected in the changes in tasks and resources. In the actual operation of the system,

the attributes of tasks, such as number, type, and priority, may change dynamically, and resources will also change dynamically. Dynamic changes in the network environment have a significant impact on data transmission, resource allocation, and task execution, which manifest in aspects such as changes in the network topology structure, vehicle mobility, and dynamic fluctuations in network load. The system performance is a key indicator for measuring the running state and stability of a system. Its dynamic changes are reflected in aspects such as latency, response time, and energy consumption, which can directly reflect the real-time operation of the system. Dynamic changes in algorithm performance directly affect the system performance, including the convergence and adaptability of the algorithm. Many algorithms exhibit obvious changes in convergence speed and convergence effect under different input data and system environments.

From the proportion of papers (see Fig. 8), there are significant differences in the research attention given to the different dynamic characteristics. Research related to dynamic changes in algorithm performance occupies a dominant position. The proportions of papers on convergence changes and adaptability changes are approximately 30.25% and 15.13%, respectively, which indicates that researchers have paid great attention to the performance stability and adaptability of algorithms in complex and changeable environments. In contrast, there is relatively less research on dynamic changes in the network environment. The proportions of studies on topology structure changes and load changes are approximately 5.04% and 3.36%, respectively, and there is still much room for improvement in this research field. In terms of dynamic changes in system performance, the proportions of papers on changes in time indicators and energy consumption changes are approximately 5.04% and 7.56%, respectively. Despite certain achievements, there is still potential for further exploration. In terms of dynamic changes in task resources, the proportions of papers on task attribute dynamic changes and system resource dynamic changes are 32.77% and 27.73%, respectively, with a relatively high total proportion. However, there is still much work to be done to understand the dynamic correlation mechanism between tasks and resources and to achieve accurate and efficient task scheduling and resource allocation.

VI. FUTURE RESEARCH DIRECTIONS

From the previous analysis of HVNet TODM algorithms, although there have been achievements in this field, there are still some deficiencies. To promote its development and meet practical needs, future research can be conducted in the following key directions, as shown in Fig. 9.

A. IMPROVE THE PERFORMANCE INDICATOR SYSTEM

The current use of performance indicators in HVNet TODM research faces several issues. Task execution latency and energy consumption have received significant attention, whereas indicators such as resource utilization and fairness are under-studied. Measuring resource utilization is difficult,

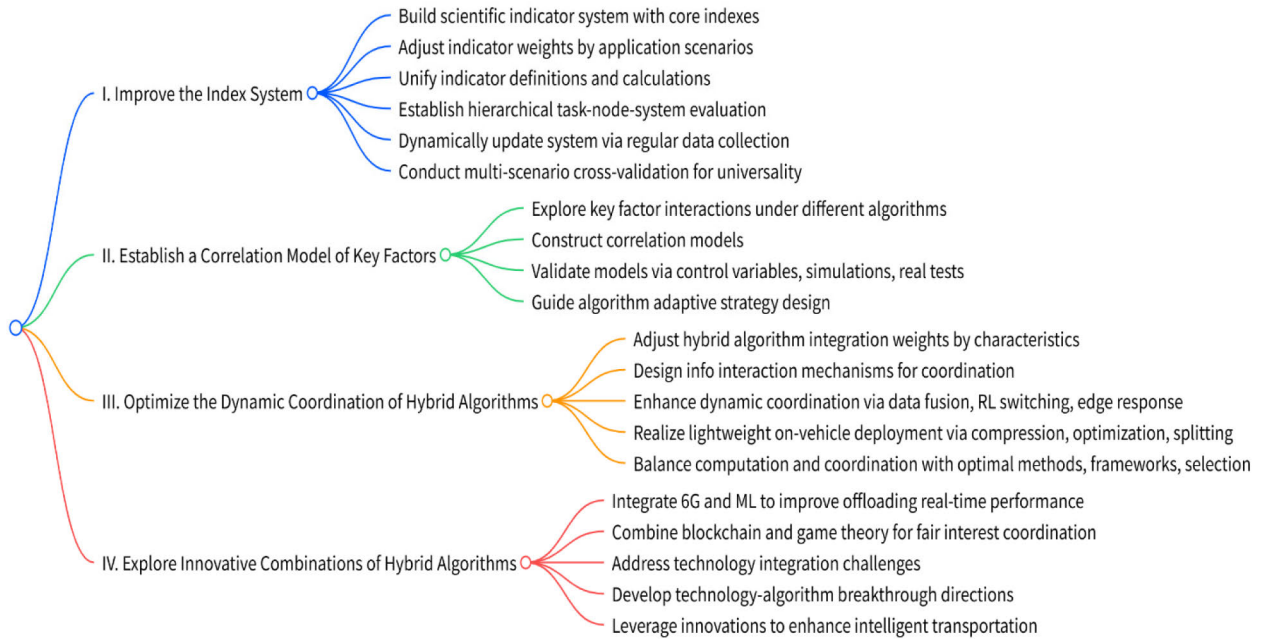


FIGURE 9. Proposed future research directions for HVNet task offloading decision-Making.

and quantifying the fairness indicator also presents challenges.

Future work should establish a performance indicators system that is both theoretically sound and widely accepted in industry, while also exploring its applicability and limitations. The algorithm taxonomy framework used in this study includes five types of algorithms. Different algorithms have different impacts on the performance indicators, which is helpful for building a more scientific system. The core indicators include the task completion rate, execution latency, energy consumption, resource utilization, system throughput, and system utility.

Traditional optimization algorithms can improve the task completion rate and resource utilization in simple and stable scenarios; however, their performance is affected in complex scenarios. Machine learning algorithms rely on data quality. Good-quality data have a positive impact on relevant indicators, whereas poor-quality data have the opposite effect. Heuristic algorithms can improve the task completion rate and system throughput in scenarios with high time requirements and relatively low complexities. However, they are prone to falling into local optima in complex scenarios, resulting in poor performance of the indicators. Game theory-driven algorithms can optimize resource allocation in scenarios with multiple participants; however, the high convergence and coordination costs in large-scale scenarios affect the task completion rate and execution latency. To enhance adaptability further, hybrid algorithms combine the advantages of multiple algorithms. For example, when

traditional optimization and machine learning algorithms are combined, the integration effect can be the key focus when designing indicators. This taxonomy framework can clarify the advantages and disadvantages of algorithms and provide new ideas for research.

Different application scenarios have different effects on performance indicators. In the intelligent transportation scenario, more attention is paid to task execution latency and completion rate, whereas in the logistics distribution scenario assisted by HVNet, energy consumption and resource utilization are more important.

We propose the following to standardize performance indicators:

- **Clearly define core indicators and their weights:** determine the core indicators and assign weights according to the requirements of different scenarios. For example, in the intelligent transportation scenario, the weights of task execution latency and completion rate are increased, and in the logistics distribution scenario, the weights of energy consumption and resource utilization are increased.
- **Unify the definitions and calculation methods of indicators:** provide clear definitions for each core indicator, and unify the calculation methods. For example, task execution latency is measured in milliseconds, and resource utilization is calculated as the percentage of resources used to total resources to ensure data comparability.

- **Establish a hierarchical evaluation mechanism:** conduct evaluations at the task, node, and system levels. At the task level, focus on the completion of individual tasks; at the node level, evaluate the resource utilization and energy consumption of each node; at the system level, consider the overall system indicators to comprehensively reflect the algorithm performance.
- **Dynamically update the evaluation system:** Owing to the dynamic changes in the Internet of Vehicles environment, collect data regularly and adjust the indicator weights and evaluation criteria according to changes in the network, tasks, and resources. For example, the weights of relevant indicators are re-evaluated when the network topology changes.
- **Conduct cross-validation in multiple scenarios:** test the algorithm in different scenarios, such as urban congestion, highway, and intelligent logistics park scenarios, to verify the universality of the evaluation system and improve the problems in a timely manner.

B. ESTABLISHMENT A CORRELATION MODEL OF KEY FACTORS

As analyzed previously, current research on various key factors is unbalanced, and correlation analysis is generally lacking. Computing resources receive high attention, but their relationships with other factors are complex, and research on storage resources, network heterogeneity, and vehicle mobility is relatively scarce, yet they have a significant impact on algorithm performance. Although task types are important, the influence mechanism between them and other factors has not been fully explored. These issues limit the performance of algorithms in actual scenarios.

We can establish a correlation model of key factors by analyzing the interactions among computing resources, network heterogeneity, and task types. This model improves algorithm performance by enabling dynamic resource allocation. The specific suggestions are as follows.

1) EXPLORE THE CORRELATION MECHANISM

The hybrid algorithm taxonomy framework is used to explore the interactions between the key factors under different algorithms. Analyze the relationships between game theory-driven algorithms and vehicle mobility, task types, connections between heuristic algorithms and computing resources, and network heterogeneity. For example, clarify how network heterogeneity affects the efficiency of computing resources, how vehicle mobility changes the demand for computing resources allocation, and the different resource demands of different task types when network heterogeneity and vehicle mobility change.

2) CONSTRUCT THE CORRELATION MODEL

- **System dynamics model:** System dynamics is used to construct a factor relationship diagram to show the causal feedback relationships among key factors, which provides support for accurate modeling. Through this

diagram, the mutual influence paths of each factor can be intuitively presented, facilitating a subsequent in-depth analysis.

- **Multi-factor integration model:** This model combines the advantages of multiple modeling methods, such as a CNN with multiple linear regression. A CNN is used to extract complex data features, and multiple linear regression is used to establish linear relationships among factors. During training, regularization techniques are adopted to find the optimal combination of hyperparameters through cross-validation, and transfer learning was used to improve the model's adaptability.
- **Correlation model based on Machine Learning:** with the help of machine learning algorithms, such as DNN, for modeling. Taking key factors as input features and training with a large amount of actual data, the model learns the interactions among factors. Combined with the attention mechanism, to make the model pay more attention to the relationships of important factors, improving performance and interpretability.

3) EXPERIMENTAL VERIFICATION OF THE MODEL

- **Experiments using the control variable method:** Use the control variable method, change one key factor each time, and observe the changes in other factors and the algorithm performance. First, we reveal the direct influence relationships, and then study the interactions when multiple factors change simultaneously.
- **Multi-scenario simulation experiments:** Construct multiple HVNet scenarios for simulation experiments, covering different task types, network environments, and vehicle movement patterns. Different combinations of key factors are set, the performance of the algorithms is recorded, and the generalization ability of the model is improved.
- **Real-scenario testing experiments:** Conduct testing experiments in an actual HVNet environment, obtain real data, verify the effectiveness of the model in the real environment, and optimize the model according to the actual situation.
- **Data-driven experimental optimization:** The experimental scheme is optimized according to the collected data. Through data analysis, exploring potential patterns and anomalies, adjusting the experimental parameter settings and scenario design, conducting in-depth analysis of the abnormal algorithm performance under specific factor combinations, and deepening the understanding of the correlations among factors.

4) THE APPLICATION PROSPECTS OF THE MODEL AND ITS IMPACT ON ALGORITHM DESIGN

- **Application prospects:** The constructed correlation model can more accurately predict changes in the HVNet environment, providing a reliable basis for TODM. In fields such as intelligent transportation and logistics distribution, it can optimize resource allocation,

improve system efficiency, and reduce congestion and energy consumption.

- **Impact on algorithm design:** Help algorithm designers comprehensively consider the effects of various factors and design algorithms that are more adaptable to complex environments. Algorithms can automatically adjust strategies according to different factor combinations based on a correlation model, thereby improving adaptability and robustness. For example, in scenarios with high vehicle mobility and network heterogeneity, the algorithm can adjust the task offloading strategy in advance to avoid task failure.

C. OPTIMIZE THE DYNAMIC COORDINATION OF HYBRID ALGORITHMS

Although hybrid algorithms integrate the advantages of multiple algorithms in the field of HVNet TODM, the problem of coordination among algorithms restricts the improvement in their performance. To address these issues, the hybrid algorithm taxonomy framework can be leveraged to enhance dynamic coordination in multiple ways.

1) A NEW PERSPECTIVE ON OPTIMIZING THE INTEGRATION STRATEGY

The hybrid algorithm taxonomy framework offers a new direction for optimizing integration strategies in future research. Different types of algorithms have their own characteristics and their integration methods and weight settings should also be different. The taxonomy framework clearly shows the characteristics of various algorithms, and researchers can use big data analysis and intelligent optimization algorithms to determine the integration strategy more accurately.

For example, in the case of a hybrid of machine learning and heuristic algorithms, the integration ratio can be dynamically adjusted according to the advantages of the two in the framework when the tasks are complex and changeable. Machine learning algorithms have advantages in handling complex data and adjusting strategies adaptively, whereas heuristic algorithms can quickly search for near-optimal solutions based on experience. When the task complexity increases, the weight of machine learning algorithms can be increased, and when the task has high requirements for response speed, the weight of heuristic algorithms can be increased to improve algorithm coordination and provide a new direction for future research on the optimization of the integration strategy.

2) NEW IDEAS FOR IMPROVING THE COORDINATION MECHANISM

Future work can use this taxonomy framework as a starting point for improving algorithm coordination mechanisms. Different types of algorithms have different operational logic in TODM, which leads to poor information interaction. The taxonomy framework helps to identify these differences and then constructs an information interaction mechanism.

For example, in the scenario of a hybrid of game theory and machine learning algorithms, a more efficient information interaction method is designed according to the differences between the TODM methods of the two in the framework. Game theory-driven algorithms focus on the strategic interaction and interest distribution among participants, whereas machine learning algorithms rely on data learning to optimize decisions. Based on this, an information interaction mechanism based on message queues and distributed shared memory can be designed to achieve asynchronous transmission and real-time sharing of task information between the two, thereby improving the algorithm coordination efficiency.

3) ENHANCE THE DYNAMIC COORDINATION ABILITY OF ALGORITHMS

Develop a multi-source data fusion collection algorithm, use Kalman filtering technology to improve data quality, and adopt autoencoders to improve data processing efficiency. In scenarios where vehicle mobility and network topology change dynamically, the multisource data fusion collection algorithm integrates multisource data, Kalman filtering removes noise, and autoencoders extract key features, providing accurate and timely data support for the algorithm, enabling the algorithm to quickly adapt to environmental changes.

Deep reinforcement learning is used to train the switching strategy model and combine it with transfer learning to enhance the generalization ability of the model. In the case of a hybrid of multiple algorithms, deep reinforcement learning enables the algorithm to learn the optimal switching strategy in different scenarios, and transfer learning applies the existing experience to new scenarios, thereby improving the adaptability and coordination of the algorithm in complex dynamic environments.

For extreme scenarios, such as sudden network congestion and equipment failures, establish an emergency response mechanism and a fast weight adjustment algorithm, and combine edge computing to improve the response speed. When extreme scenarios occur, the emergency response mechanism is quickly activated, and edge computing devices process local data in a timely manner, quickly adjusting the algorithm weights to ensure stable performance of the algorithm.

4) EXPLORE THE FEASIBILITY OF DEPLOYING LIGHTWEIGHT HYBRID ALGORITHMS AT THE VEHICLE END

The limited resources of vehicle-end devices in HVNet make it essential to explore the deployment of lightweight hybrid algorithms. Currently, computing power, storage capacity, and energy of vehicle-end devices are limited. For example, common vehicle-mounted chips have a small number of computing cores, low main frequency, and relatively small memory and storage capacity. Moreover, the vehicle-mounted energy mainly depends on the battery, and battery life affects the running time of the device.

This makes it difficult to directly deploy traditional hybrid algorithms. Complex hybrid algorithms have high computa-

tional complexity, which makes it difficult for vehicle-end devices to bear, resulting in an increase in task-processing latency and an inability to meet real-time requirements. Simultaneously, the large memory occupation during the operation of the algorithm may exceed the memory capacity of vehicle-end devices, causing the system to freeze or even crash. In addition, high-energy-consuming algorithms accelerate battery power consumption and affect the overall battery life of the vehicle.

To deploy lightweight hybrid algorithms at the vehicle end, multiple aspects can be started. On the one hand, adopt model compression technologies, such as pruning and quantization processing of machine learning models. Removing unimportant connections and parameters in the model through pruning and quantization converts high-precision data into low-precision data, reducing computational complexity and memory occupation while ensuring a small loss of algorithm accuracy. However, the algorithm structure is optimized. For example, replace complex deep learning models with lightweight models, such as architectures designed specifically for resource-constrained devices, such as MobileNet and ShuffleNet. Simultaneously, we explore strategies for reasonably allocating tasks between the vehicle end and ES. The vehicle end processes simple parts with high real-time requirements, and complex computing tasks are handed over to the ES, thereby reducing the computational pressure at the vehicle end and improving the overall performance of the algorithm.

5) BALANCE THE RELATIONSHIP BETWEEN COMPUTATION AND COORDINATION

In the process of optimizing the dynamic coordination of hybrid algorithms, computational trade-offs are a key link that cannot be ignored. This optimization aims to reduce computational complexity and enhance inter-algorithm coordination efficiency without compromising overall performance.

Exploring new near-optimal calculation methods and lightweight algorithms: Researching new near-optimal calculation methods and lightweight algorithms is important for balancing the relationship between computation and coordination. In the scenario of a hybrid of algorithms of the same type, for example, in the application of the particle swarm optimization algorithm and its variants, we deeply explore efficient algorithm improvement schemes. The number of calculation steps can be reduced by optimizing the search strategy and parameter settings of the algorithm. To ensure that the algorithm can find high-quality near-optimal solutions, reduce computational complexity, and avoid affecting the coordination effect among algorithms owing to the excessive computational burden.

Develop a distributed computing framework for the coordination of edge and cloud computing: Developing a distributed computing framework for the coordination of edge and cloud computing can optimize the allocation of computing resources and improve overall computing efficiency. Complex computing tasks are reasonably split and allocated to

edge devices and cloud servers. Edge devices, which have the advantage of low latency, process parts of tasks with high real-time requirements; cloud servers use their powerful computing capabilities to undertake large-scale data processing and complex computing tasks. In this way, not only can the overall computational complexity be reduced, but more favorable conditions can also be created for algorithm coordination, ensuring that each algorithm runs efficiently in different computing resources environments and improves the overall performance of hybrid algorithms.

Adopt an adaptive algorithm selection strategy: During the algorithm design stage, an adaptive algorithm selection strategy, when combined with meta-learning technology, can better balance convergence speed, solution quality, and algorithmic complexity. According to the real-time situation of tasks and resources, such as the urgency of tasks and computational requirements, as well as the available computing resources and network conditions in HVNet, appropriate algorithms or algorithm combinations are adaptively selected. Meta-learning technology helps the algorithm quickly learn the optimal TODM patterns in different scenarios, improving the accuracy and efficiency of algorithm selection, ensuring that the algorithm achieves the best performance when working in coordination, and effectively coping with the complex dynamic environment of HVNet.

D. EXPLORE INNOVATIVE COMBINATIONS OF HYBRID ALGORITHMS

The ongoing advancement of emerging technologies—such as 6G, blockchain, and quantum computing—presents significant opportunities for enhancing HVNet TODM algorithms. The hybrid algorithm taxonomy framework established in this survey provides a structured foundation for exploring these innovations. It offers a principled approach to guiding the integration of emerging technologies with existing algorithmic paradigms, thereby enabling more systematic and effective future research in this domain.

1) INSPIRATION FOR THE INTEGRATION DIRECTION

The hybrid algorithm taxonomy framework can help future research determine how well emerging technologies adapt to different algorithms, thereby identifying new integration directions. The framework clearly presents the advantages and disadvantages of various algorithms, which is helpful for analyzing the feasibility of combining emerging technologies with existing algorithms.

Consider 6G technology as an example. Its high-speed and low-latency characteristics have great potential for combination with machine learning algorithms. Machine learning algorithms rely on a large amount of data for learning, and TODM and 6G technology can provide timely and abundant data support. In the TODM scenario, the high-speed communication capability of 6G allows machine learning algorithms to quickly obtain data, such as the vehicle's surrounding environment and task requirements, process them rapidly and make decisions, and significantly improve the real-time

performance and accuracy of TODM. This lays the foundation for subsequent research into integrating 6G technology with machine learning algorithms. For example, it can explore how to optimize the data transmission protocol to ensure that data are efficiently and stably transmitted to machine learning algorithms under the 6G network and study how to improve the model structure of machine learning algorithms to better adapt to the large amount of data and rapid data updates in the 6G environment.

2) A NEW PERSPECTIVE FOR TAPPING POTENTIAL

The hybrid algorithm taxonomy framework also provides a novel perspective for exploring the full potential of integrating emerging technologies with existing algorithms. The integration of blockchain technology with game-theoretic algorithms serves as a compelling example. Game theory-driven algorithms are often used to analyze strategic interaction and interest distribution relationships among multiple agents. The decentralized and tamper-proof characteristics of blockchain provide new ideas for the application of game theory-driven algorithms to task offloading in HVNet.

In the TODM process, multiple agents, such as vehicles and ES, are involved, and there are complex interest relationships. With the help of the distributed ledger technology of blockchain, the task offloading behaviors, resource usage situations and other information of each agent can be accurately recorded, and this information is tamper-proof, ensuring the authenticity and reliability of the data. This provides a more reliable TODM basis for game theory-driven algorithms, making the TODM process of each agent fairer and more transparent, and effectively avoiding the interference of malicious nodes. Subsequent research can start from this perspective, further explore how to construct a more reasonable incentive mechanism based on blockchain technology to encourage each agent to actively participate in cooperation, and study how to optimize the smart contracts of blockchain to better combine with game-theory algorithms and achieve more efficient TODM.

However, integrating emerging technologies with existing task offloading algorithms presents significant challenges. 6G technology has problems in terms of network coverage and equipment costs. The data storage and smart contract execution efficiency of the blockchain technology must be improved. Quantum computing technology has a low maturity and high equipment costs.

Future research should focus on the characteristics of emerging technologies and the actual needs of task offloading in HVNet. Deeply studying the interaction mechanism between 6G communication and existing algorithms, taps the potential of blockchain to ensure the security and trust of task offloading, and explores more effective integration methods between quantum computing and traditional algorithms. Simultaneously, factors such as computational efficiency, energy consumption, and security should be comprehensively considered, and integrated algorithms should be

systematically optimized to construct a comprehensive and efficient TODM system.

In terms of breakthrough directions, quantum computing technology is expected to progress, achieving deep integration of more powerful quantum algorithms and traditional algorithms to solve the problem of ultra-large-scale task offloading. By combining blockchain encryption technology and 6G high-speed communication, a secure and efficient distributed computing platform for HVNet can be constructed.

In terms of the impact on the intelligent transportation field, the integration of emerging technologies and task offloading algorithms will promote the development of intelligent transportation systems for intelligence, efficiency, and security. It can reduce the computational burden of vehicles, improve the response speed of in-vehicle systems, enhance driving safety and comfort, promote the deep integration of HVNet with other components of intelligent transportation systems, improve the overall operation efficiency of the transportation system, and reduce traffic congestion and energy consumption.

In conclusion, the integration of emerging technologies and HVNet TODM algorithms is full of opportunities and challenges. Continuous research and innovation are expected to drive significant advancements in intelligent transportation, meeting its growing demands and fostering ongoing industry development.

VII. CONCLUSION

This study has presented a comprehensive survey of HVNet TODM algorithms. Through a review of papers from the past decade, this study makes several key contributions: it proposes a new taxonomy framework for hybrid algorithms, explores correlations among key factors, constructs a computational trade-off framework, and emphasizes the need for performance indicator standardization. The main contributions are as follows: innovating the algorithm taxonomy, subdividing hybrid algorithms, providing a new perspective for research, analyzing key factors and their correlations, studying the characteristics of algorithms, providing a basis for algorithm design and selection, analyzing the dynamic performance of algorithms, and proposing improvement directions.

However, there are still some limitations in this field. In terms of performance indicators, the degree of attention paid to each indicator varies greatly, some indicators are understudied and lack unified standards; in the research of key factors, the research on each factor is unbalanced, and the correlation analysis is lacking; hybrid algorithms have problems in dynamic coordination. Future research should focus on several areas: (1) enhancing the performance indicators system by deepening the study of under-represented performance indicators like resource utilization and establishing unified standards; (2) developing correlation models for key factors to improve algorithm adaptability, exploring the correlations among factors, improving the adaptability of algorithms, optimizing the dynamic characteristics of

hybrid algorithms, improving the integration and coordination mechanisms, combining emerging technologies such as 6G, exploring innovative combinations of hybrid algorithms, and improving the intelligence level of TODM.

Research on HVNet TODM algorithms holds significant practical importance. The new framework has significant value in emerging fields such as intelligent logistics, integration of HVNet, and the Industrial Internet of Things. It can optimize task allocation, improve efficiency, and ensure data security. Improving the algorithm system enhances the performance of the HVNet system and promotes the development of intelligent transportation systems.

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